

The Square-Root Horizon: Squares, Reversal and Prime Clocks in the Distribution of the Primes and the Zeros of the Riemann Zeta Function

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Abstract

We read the primes and the zeros of the Riemann zeta function through three exact structures: the residue classes of the divisor in the rebuild law $\sum_{r \leq x} M(x/r) = 1$, the squares of the primes, and the reversal of the Euler product. Split by $r \bmod 3$ the rebuild law is $(1, a(x), -a(x))$. Its zero class reproduces the whole law at $x/3$, so the classes modulo 3^k form a self-similar tower whose leaf energy is $\ll x^{1+\varepsilon}$ for every $\varepsilon > 0$ if and only if the Riemann Hypothesis (RH) holds; at every zero of ζ the three classes stand as $(0, a, -a)$. Each prime square enters the logarithm of the Euler product with weight $\frac{1}{2}$, and at that weight the lean of the prime count vanishes: the weight fitted from the exact counts to 10^{10} is 0.4994. The squares part of ζ is zero-free for $\sigma > \frac{1}{2}$ and has a half-order pole at $s = \frac{1}{2}$, the square-root horizon, matched by a half-order zero of the prime factor, which carries every zero of ζ in $\sigma > \frac{1}{2}$. Reversing the sign of every prime turns $\zeta(s)$ into $\zeta(2s)/\zeta(s)$ and every zero into a pole; the Liouville-weighted sawtooths rebuild the square staircase exactly, and if they approximate the odd-square staircase in $L^2(0, \infty)$ then RH holds. On the critical line every forward term of the approximate functional equation is balanced by its reversed partner, and off it no two pairs are balanced at once; the pair truncated without its remainder has zeros off the line, at each of which ζ itself is far from zero. The Jensen polynomials of Ξ , read as Jacobi chains, have every coupling positive to degree 400, and each prime pulls the zeros toward the reversal of its clock, as Landau's formula prescribes. We state the axiom of harmonic independence: the prime waves are related only through the mechanism of their construction, and a zero off the critical line would be a relation without a mechanism. The Davenport–Heilbronn function, whose zeros leave the line, adds two Euler products with the root number as weight and couples distinct primes in its logarithm, as the axiom predicts.

Keywords: Euler product; Mertens function; Liouville function; Riemann Zeta function; prime number theory

1. Introduction

Let μ be the Möbius function, λ the Liouville function and Λ the von Mangoldt function, and let $M(x) = \sum_{n \leq x} \mu(n)$, $L(x) = \sum_{n \leq x} \lambda(n)$, $\pi(x)$, $\theta(x) = \sum_{p \leq x} \log p$, $\psi(x) = \sum_{n \leq x} \Lambda(n)$ and $J(x) = \sum_{k \geq 1} \pi(x^{1/k})/k$; functions of x vanish for $x < 1$. We write $\pi(x; q, a)$, $\theta(x; q, a)$ and $\psi(x; q, a)$ for the same counts restricted to $n \equiv a \pmod{q}$, $\text{li}(x)$ for the logarithmic integral, χ_{-3} and χ_{-4} for the nonprincipal characters modulo 3 and 4, $L(s, \chi)$ for Dirichlet L -functions, $\zeta(s, a)$ for the Hurwitz zeta function and $\eta(s) = \sum_{n \geq 1} (-1)^{n-1} n^{-s} = (1 - 2^{1-s})\zeta(s)$; $\nu(n)$ and $\Omega(n)$ are the numbers of distinct and of all prime factors of n , and γ_E is Euler's constant. The nontrivial zeros of ζ are $\rho = \beta + i\gamma$, counted with multiplicity; $N(T)$ is the number with $0 < \gamma \leq T$. The Riemann Hypothesis (RH) is the statement that $\beta = \frac{1}{2}$ for every ρ . All zeros with $0 < \gamma \leq 3 \cdot 10^{12}$ lie on the line $\sigma = \frac{1}{2}$ [46]. Hardy proved that infinitely many zeros lie on the line [29], Selberg that a positive proportion do [55], Levinson more than one third [34], Conrey more than two fifths [19], Bui, Conrey and Young more than 41% [14], and Pratt, Robles, Zaharescu and Zeindler more than five twelfths [49]. Montgomery's pair-correlation method [39], made independent of RH by Aryan [3] and by Baluyot, Goldston, Suriajaya

and Turnage-Butterbaugh [7], shows that at least two thirds of the zeros are simple and on the line if all zeros with $T < \gamma \leq 2T$ lie within $0.3185/\log T$ of it [8]. In 2026 two results obtained by AI models were reported. Anthropic reported an unconditional proof, with a Lean formalisation, that at least 67.2% of the zeros, counted with multiplicity, are simple and on the line, against the previous record $\frac{5}{12}$ of [49] (given there as 41.6%); the argument combines the unconditional pair-correlation theorems of [3,7] with Bombieri's work on Weil's quadratic functional [13], and extends to primitive Dirichlet L -functions [4,18]. OpenAI released a manuscript asserting that every Dirichlet L -function, ζ included, and every finite-order Hecke L -function over $\mathbb{Q}(\sqrt{-3})$ has no zeros with $\operatorname{Re} s > \frac{7}{8}$ other than the pole at $s = 1$, with a Lean formalisation of these statements [44]; a companion manuscript gives a different proof of the half-plane $\operatorname{Re} s > \frac{11}{12}$ [45]. The de Bruijn–Newman constant Λ_{dBN} satisfies $0 \leq \Lambda_{\text{dBN}} \leq 0.2$ [22,40,48,52], [46, Corollary 2], and RH is equivalent to $\Lambda_{\text{dBN}} = 0$. We measure x in doublings, $u = \log_2 x$; a zero $\frac{1}{2} + i\gamma$ contributes to the counts a wave $x^{i\gamma} = e^{2\pi i u \gamma \log 2/2\pi}$ whose pitch is $\gamma \log 2/2\pi$ turns per doubling.

Pairing the squarefree divisors $\{d, dp\}$ of every integer $k > 1$ gives $\sum_{d|k} \mu(d) = 0$, and summing over $k \leq x$ gives the rebuild law [1, Ch. 3]

$$\sum_{r \leq x} M(x/r) = 1 \quad (x \geq 1), \quad (1)$$

by which the balance of the Möbius sum at every scale is fixed by the balances at all smaller scales. The present paper reads the law, the prime counts and ζ itself through three exact structures: the residue classes of the divisor r in (1); the squares of the primes, which sit on the square-root horizon $\sigma = \frac{1}{2}$ where the series $\sum_p p^{-2\sigma}$ stops converging; and the reversal $p \mapsto -p$ of the Euler product, which turns every zero of ζ into a pole.

1.1. Main Results

For $m \geq 1$, a residue $c \bmod m$ and a character $\chi \bmod m$ put

$$S_c^{(m)}(x) = \sum_{\substack{r \leq x \\ r \equiv c \pmod{m}}} M(x/r), \quad G_\chi(x) = \sum_{r \leq x} \chi(r) M(x/r) = \sum_{k \leq x} (\mu * \chi)(k), \quad (2)$$

and write $G_{-3} = G_{\chi_{-3}}$, $G_{-4} = G_{\chi_{-4}}$.

- Theorem 1** (the triad tower). (i) For $x \geq 3$, $(S_0^{(3)}, S_1^{(3)}, S_2^{(3)})(x) = (1, a(x), -a(x))$ with $a(x) = \frac{1}{2}G_{-3}(x)$; for $x \geq 4$, $(S_0^{(4)}, S_1^{(4)}, S_2^{(4)}, S_3^{(4)})(x) = (1, b(x), 0, -b(x))$ with $b(x) = \frac{1}{2}G_{-4}(x)$.
(ii) For $\kappa \geq 0$, every c and every $x > 0$, $S_{3c}^{(3^\kappa)}(x) = S_c^{(3^\kappa)}(x/3)$; for a unit $v \bmod 3^\kappa$, $S_v^{(3^\kappa)}(x) = \varphi(3^\kappa)^{-1} \sum_{\chi \bmod 3^\kappa} \bar{\chi}(v) G_\chi(x)$.
(iii) The energy $\mathcal{E}_\kappa(x) = \sum_{c \bmod 3^\kappa} S_c^{(3^\kappa)}(x)^2$ of level κ satisfies

$$\mathcal{E}_\kappa(x) = \mathcal{E}_{\kappa-1}(x/3) + \frac{1}{\varphi(3^\kappa)} \sum_{\chi \bmod 3^\kappa} |G_\chi(x)|^2 \quad (\kappa \geq 1). \quad (3)$$

- (iv) If $3^\kappa > x$ the classes are the leaves $S_r^{(3^\kappa)}(x) = M(x/r)$, and $\mathcal{E}_\infty(x) = \sum_{r \leq x} M(x/r)^2$. RH holds if and only if $\mathcal{E}_\infty(x) \ll_\varepsilon x^{1+\varepsilon}$ for every $\varepsilon > 0$.

Theorem 2 (the carriers). For every character χ , prime p and $e \geq 1$, $(\mu * \chi)(p^e) = \chi(p)^{e-1}(\chi(p) - 1)$. Hence $\mu * \chi$ vanishes on every multiple of a prime p with $\chi(p) = 1$; for $\chi \bmod m$ this includes every prime $p \equiv 1 \pmod{m}$. In particular $\mu * \chi_{-3}$ is supported on the integers $3^j n$ with $j \in \{0, 1\}$ and n composed of primes $\equiv 2 \pmod{3}$, where $(\mu * \chi_{-3})(3^j n) = (-1)^{j+\Omega(n)} 2^{v(n)}$.

Theorem 3 (the triad at a zero). Let $A_c(s) = \sum_{n \equiv c(3)} n^{-s}$ and $B_c(s) = \sum_{n \equiv c(4)} n^{-s}$. Then $A_0 = 3^{-s}\zeta(s)$, $A_c = 3^{-s}\zeta(s, c/3)$ for $c = 1, 2$, $A_1 + A_2 = (1 - 3^{-s})\zeta(s)$ and $A_1 - A_2 = L(s, \chi_{-3})$, and at every nontrivial zero ρ of ζ

$$\begin{aligned}(A_0, A_1, A_2)(\rho) &= (0, \tfrac{1}{2}L(\rho, \chi_{-3}), -\tfrac{1}{2}L(\rho, \chi_{-3})), \\ (B_0, B_1, B_2, B_3)(\rho) &= (0, \tfrac{1}{2}L(\rho, \chi_{-4}), 0, -\tfrac{1}{2}L(\rho, \chi_{-4})).\end{aligned}$$

The Dirichlet series whose summatory function is $S_c^{(m)}$ is $m^{-s}\zeta(s, c/m)/\zeta(s)$ ($c \neq 0$), and at a simple zero ρ the residue of $m^{-s}\zeta(s, c/m)x^s/(s\zeta(s))$, the coefficient of the wave x^ρ of $S_c^{(m)}(x)$, is $m^{-\rho}\zeta(\rho, c/m)x^\rho/(\rho\zeta'(\rho))$.

Theorem 4 (primes, squares and wheels). (i) (window law) For consecutive primes $p < p'$, the integers in $[p, pp']$ with no prime factor below p are exactly the primes in $[p, pp']$ together with p^2 .
(ii) (squares land in one class) For every prime $p \geq 5$, $p^2 \equiv 1 \pmod{24}$.
(iii) (crossing law) In any increasing sequence of integers $\equiv \pm 1 \pmod{6}$, in particular the primes ≥ 5 with or without the prime squares inserted, the number of gaps $\equiv 2 \pmod{6}$ and the number of gaps $\equiv 4 \pmod{6}$ differ by at most 1.

Theorem 5 (the logarithm of the wheel). Let W be squarefree.

- (i) For $\sigma > 1$, $\sum_{(n,W)=1} n^{-s} = \zeta(s) \prod_{q|W} (1 - q^{-s})$, and every zero of ζ with $\sigma > 0$ is a zero of this function of the same order. The count $\Phi_W(x) = \#\{n \leq x : (n, W) = 1\}$ satisfies $\Phi_W(x) - x\varphi(W)/W = -\sum_{d|W} \mu(d)\{x/d\}$, periodic with period W and bounded by $2^{v(W)}$.
- (ii) $\log \zeta(s) = \sum_p \sum_{k \geq 1} p^{-ks}/k = \int_1^\infty x^{-s} dJ(x)$ for $\sigma > 1$: each prime square enters with weight $\frac{1}{2}$, and $\pi(x) = \sum_{k \geq 1} \mu(k)J(x^{1/k})/k$.
- (iii) $\pi(x) - \text{li}(x) = (J(x) - \text{li}(x) + \log 2) - \frac{1}{2} \text{li}(\sqrt{x}) + O(\sqrt{x} e^{-c\sqrt{\log x}})$, and $J(x) - \text{li}(x) + \log 2 = -\sum_\rho \text{li}(x^\rho) + \int_x^\infty \frac{dt}{t(t^2-1)\log t}$ for $x > 1$, with J taken at its jumps as the mean of its one-sided limits.
- (iv) $L(x) = \sum_{d \leq \sqrt{x}} M(x/d^2)$.
- (v) $\psi(x; 6, 1) - \psi(x; 6, 5) - (\theta(x; 6, 1) - \theta(x; 6, 5)) = \theta(\sqrt{x}) - \log 6 + O(x^{1/3} \log x)$ for $x \geq 9$.

Theorem 6 (the square-root horizon). Let $S(s) = \sum_p \sum_{k \geq 2} p^{-ks}/k$, $E = e^S$ and $F = \zeta/E$.

- (i) S converges absolutely for $\sigma > \frac{1}{2}$, where $S(s) = -\sum_{k \geq 2} \mu(k)k^{-1} \log \zeta(ks)$; E is analytic and zero-free there; F is meromorphic in $\sigma > \frac{1}{2}$, equals $\exp \sum_p p^{-s}$ for $\sigma > 1$, and has exactly the zeros of ζ , with the same orders.
- (ii) For $\sigma > \frac{1}{2}$ and all real t , $E(\sigma)^3 |E(\sigma + it)|^4 |E(\sigma + 2it)| \geq 1$.
- (iii) $E(s)^2 = \zeta(2s) \exp(-2 \sum_{k \geq 3} \mu(k)k^{-1} \log \zeta(ks))$ extends meromorphically to $\sigma > \frac{1}{3}$; on $\sigma \geq \frac{1}{2}$ its only singularity is a simple pole at $s = \frac{1}{2}$ and it has no zeros there. Hence $F(s)^2$ has a simple zero at $s = \frac{1}{2}$, and

$$\log E(\sigma) + \tfrac{1}{2} \log(2\sigma - 1) \rightarrow C_E, \quad \log |F(\sigma)| - \tfrac{1}{2} \log(2\sigma - 1) \rightarrow \log |\zeta(\tfrac{1}{2})| - C_E \quad (\sigma \downarrow \tfrac{1}{2}),$$

with $C_E = -\sum_{k \geq 3} \mu(k)k^{-1} \log \zeta(k/2) = 0.36399$ and $\log |\zeta(\frac{1}{2})| - C_E = 0.01469$.

- (iv) At every zero $\rho = \frac{1}{2} + i\gamma$ the squared pair $Q_\zeta(\sigma, t) = 3 \log |\zeta(\sigma)| + 4 \log |\zeta(\sigma + it)| + \log |\zeta(\sigma + 2it)|$ tends to $-\infty$ as $\sigma \rightarrow \frac{1}{2} +$ with $t = \gamma$, and so does the squared pair Q_F of the prime factor.

Theorem 7 (the prime term at every height). Let (ϑ_p) be independent and uniform on $[0, 2\pi)$ and $\sigma > 0$.

- (i) The flow $t \mapsto (t \log p / 2\pi \bmod 1)_{p \leq y}$ is equidistributed on the torus $(\mathbb{R}/\mathbb{Z})^{\pi(y)}$ for every y .
- (ii) $\mathbb{E} |\sum_{p \leq y} p^{-\sigma} e^{i\vartheta_p}|^2 = \sum_{p \leq y} p^{-2\sigma}$, and $\sum_p p^{-\sigma} e^{i\vartheta_p}$ converges almost surely if and only if $\sigma > \frac{1}{2}$.
- (iii) For $\sigma > \frac{1}{2}$ the random Euler product $Z_\sigma = \prod_p (1 - p^{-\sigma} e^{i\vartheta_p})^{-1}$ converges almost surely and in mean square, $Z_\sigma \neq 0$ almost surely, and $\mathbb{E} |Z_\sigma|^2 = \zeta(2\sigma) = \sum_n n^{-2\sigma}$.
- (iv) (Hardy and Littlewood; Bohr and Landau; Selberg; Bohr and Jessen) For fixed $\sigma > \frac{1}{2}$, $\frac{1}{T} \int_1^T |\zeta(\sigma + it)|^2 dt \rightarrow \zeta(2\sigma)$; the number $N(\sigma, T)$ of zeros with $\beta > \sigma$, $0 < \gamma \leq T$ is $O(T)$

and $O(T^{1-(\sigma-\frac{1}{2})/4} \log T)$, so that these zeros form a proportion $O(1/\log T)$ of all zeros up to height T ; and $\log \zeta(\sigma + it)$ has an absolutely continuous limiting distribution.

Theorem 8 (the reversed pattern). (i) For $\sigma > 1$, $\prod_p (1 + p^{-s})^{-1} = \zeta(2s)/\zeta(s) = \sum_n \lambda(n)n^{-s}$. Every zero of ζ with $\beta \geq \frac{1}{2}$ is a pole of $\zeta(2s)/\zeta(s)$ of the same order; its centre $s = \frac{1}{2}$ is a simple pole with residue $1/(2\zeta(\frac{1}{2}))$.

(ii) Forward times reversed is the pattern of the squares: $\zeta(s) \cdot \zeta(2s)/\zeta(s) = \zeta(2s)$, that is $\sum_{d|n} \lambda(d) = [n \text{ is a square}]$, and

$$\sum_{n \leq x} L(x/n) = \lfloor \sqrt{x} \rfloor \quad (x > 0). \quad (4)$$

(iii) $\sum_{n \geq 1} \lambda(n)/n = 0$, and for every $x > 0$ the series $\sum_{k \geq 1} \lambda(k)\{x/k\}$ converges, with

$$-\sum_{k \geq 1} \lambda(k)\{x/k\} = \lfloor \sqrt{x} \rfloor, \quad \sum_{k \leq N} \lambda(k)\{x/k\} = x \ell(N) - \lfloor \sqrt{x} \rfloor \quad (N \geq x), \quad (5)$$

where $\ell(N) = \sum_{k \leq N} \lambda(k)/k$.

For $t > 0$ let $\varsigma_k(t) = \{1/(kt)\}$, let $f(t) = 1$ if $0 < t \leq 1$ and $\lfloor t^{-1/2} \rfloor$ is odd and $f(t) = 0$ otherwise (the odd-square staircase), and let

$$D_N^2 = \min_{c \in \mathbb{R}^N} \left\| f - \sum_{k \leq N} c_k \varsigma_k \right\|_{L^2(0, \infty)}^2, \quad d_N^2 = \min_{c \in \mathbb{R}^N} \left\| \chi_{(0,1]} - \sum_{k \leq N} c_k \varsigma_k \right\|_{L^2(0, \infty)}^2.$$

Theorem 9 (the reversed staircase). (i) $\|f\|^2 = \eta(2) = \pi^2/12$ and $\int_0^\infty f(t)t^{s-1} dt = \eta(2s)/s$ for $\sigma > 0$; and for every $t > 0$

$$-\sum_{k \geq 1} (\lambda(k) - 2\lambda(k/4)[4 | k])\varsigma_k(t) = f(t),$$

the Dirichlet series of the weights being $\eta(2s)/\zeta(s)$.

(ii) If $D_N \rightarrow 0$ as $N \rightarrow \infty$, then $\zeta(s) \neq 0$ for $\sigma > \frac{1}{2}$, and every zero of ζ in the critical strip lies on the line $\sigma = \frac{1}{2}$.

(iii) For every N , every $c \in \mathbb{R}^N$ and every zero $\rho = \frac{1}{2} + i\gamma$, the Mellin transform of $f - \sum_{k \leq N} c_k \varsigma_k$ at ρ equals $\eta(2\rho)/\rho = \eta(1 + 2i\gamma)/\rho$; for the unit step it equals $1/\rho$.

Theorem 10 (prime clocks; Landau [33]). For fixed $x > 1$, $\sum_{0 < \gamma \leq T} x^\rho = -\frac{T}{2\pi} \Lambda(x) + O(\log T)$. Consequently, when the zeros up to height T lie on the line, $\frac{1}{N(T)} \sum_{0 < \gamma \leq T} \cos(\gamma \log x) = -\frac{T}{2\pi N(T)} \frac{\Lambda(x)}{\sqrt{x}} + O(\frac{\log T}{N(T)})$, and for each prime p and each $k \geq 1$ the k -th Fourier coefficient of the distribution of the readings $\gamma \log p / 2\pi \bmod 1$ of the clock $t \mapsto p^{it}$ is $-\frac{T}{2\pi N(T)} p^{-k/2} \log p + O(\frac{\log T}{N(T)})$.

1.2. Main Computations

The exact sieve to 10^{10} gives the class values of the triad tower at $x = 10^{10}$ at every level, with $M(10^{10}) = -33,722$, $G_{-3}(10^{10}) = -25,880$, $G_{-4}(10^{10}) = -66,506$ and the level-two sums in $\mathbb{Z}[\omega]$ (Table 1); the wave amplitudes of M , a and b at the first twelve zeros agree with Theorem 3 to within 0.0003, and their energies agree with the sums over the 2469 zeros below height 3000 to within 1% (Tables 3 and 4, Figure 1). The prime gaps up to 61 with 25 and 49 inserted, the balance 9 : 9 of the classes $6k \pm 1$, and the class counts to 10^8 are in Section 3. The leans of four prime counts over $2^{12} \leq x \leq 10^{10}$, with and without the squares at weight $\frac{1}{2}$, and the fitted square weight 0.4994 are in Table 7 and Figure 2. The half-order limits of the squares part and the prime factor, and the reach of the squared pair of ζ at 138 zeros up to height 3000, are in Tables 8 and 9 and Figure 3. The value distribution of $\log |\zeta(\sigma + it)|$ at 6000 heights in $[5000, 6000]$ and its deepest dips below height 3000 are in Table 10 and Figure 4. The reversal identities are checked in Table 14, and the identity at the first

five zeros in Table 13. The exact weights of the staircase, cut at N , are in Table 15. The distances D_N and d_N , computed from exact integrals for $N \leq 2000$, are in Tables 16 and 17 and Figure 7; Landau's formula and the density of the zeros on the clock of 2 are in Tables 18 and 19 and Figure 8. The forward and reversed terms, the zeros of the pair truncated without its remainder, and the Jensen chains of Ξ are in Table 11 and Figures 5 and 6.

1.3. Organisation

Section 2 proves Theorems 1–3 and gives the tower at 10^{10} . Section 3 proves Theorem 4 and gives the gaps and classes. Section 4 proves Theorem 5 and measures the leans. Section 5 proves Theorem 6. Section 6 proves Theorem 7(i)–(iii) and measures the value distribution. Section 7 states the axiom of harmonic independence, and Section 8 reads it at the midpoint: it compares the forward and reversed terms, computes the pair without its remainder and tests the Jensen polynomials of Ξ as chains. Section 9 proves Theorem 8. Section 10 proves Theorem 9 and computes the distances. Section 11 treats the prime clocks and the tenth zero. Section 12 describes the computations, Section 13 gathers the results, and Section 14 lists the external inputs.

2. The Triad Tower of the Rebuild Law

2.1. The Splits and the Tower

Proof of Theorem 1. (i) The integers $r \equiv 0 \pmod{3}$ are $r = 3r'$, so $S_0^{(3)}(x) = \sum_{r' \leq x/3} M((x/3)/r') = 1$ for $x \geq 3$ by (1) at $x/3$. By (1) at x , $S_1^{(3)} + S_2^{(3)} = 1 - S_0^{(3)} = 0$, while $S_1^{(3)} - S_2^{(3)} = \sum_r \chi_{-3}(r) M(x/r) = G_{-3}(x)$. The second expression for G_χ in (2) follows from $\sum_{rn \leq x} \chi(r) \mu(n) = \sum_{k \leq x} \sum_{rn=k} \chi(r) \mu(n)$. Modulo 4: $S_0^{(4)}(x) = 1$ for $x \geq 4$ as before; $S_0^{(4)} + S_2^{(4)} = \sum_{r \text{ even}} M(x/r) = 1$ for $x \geq 2$ by (1) at $x/2$, so $S_2^{(4)} = 0$ for $x \geq 4$; $S_1^{(4)} + S_3^{(4)} = 1 - 1 = 0$ and $S_1^{(4)} - S_3^{(4)} = G_{-4}(x)$.

(ii) The integers $r \equiv 3c \pmod{3^{k+1}}$ are $r = 3r'$ with $r' \equiv c \pmod{3^k}$, and $M(x/(3r')) = M((x/3)/r')$. For a unit v , orthogonality $\sum_{\chi \bmod q} \bar{\chi}(v) \chi(r) = \varphi(q)[r \equiv v \pmod{q}]$ gives the character formula.

(iii) Split the classes modulo 3^k into the multiples of 3, whose squares sum to $\mathcal{E}_{k-1}(x/3)$ by (ii), and the units. By (ii) and orthogonality, $\sum_v |S_v|^2 = \varphi(3^k)^{-2} \sum_{\chi, \chi'} \overline{G_\chi} G_{\chi'} \sum_v \chi(v) \bar{\chi}'(v) = \varphi(3^k)^{-1} \sum_\chi |G_\chi|^2$.

(iv) If $3^k > x$ each class contains at most one $r \leq x$. If RH holds then $M(y) \ll_\varepsilon y^{1/2+\varepsilon}$ [58, Theorem 14.25(C)], so $\mathcal{E}_\infty(x) \ll \sum_{r \leq x} (x/r)^{1+2\varepsilon} \leq \zeta(1+2\varepsilon) x^{1+2\varepsilon}$. Conversely $M(x)^2$ is the term $r = 1$ of $\mathcal{E}_\infty(x)$, so $\mathcal{E}_\infty(x) \ll x^{1+\varepsilon}$ gives $M(x) \ll x^{1/2+\varepsilon/2}$, and RH follows by the same theorem. $\square$

The principal character modulo 3^k contributes nothing to (3) once $x \geq 3$: $G_{\chi_0}(x) = \sum_{3|r} M(x/r) = 1 - S_0^{(3)}(x) = 0$. At level one the recursion reads $\mathcal{E}_1(x) = 1 + \frac{1}{2} G_{-3}(x)^2$ for $x \geq 3$. The tower therefore stores the rebuild law in a self-similar way: every zero class is the whole tower one level down at $x/3$, and each level adds only the character sums of its unit classes.

2.2. The Carriers

Proof of Theorem 2. For $e \geq 1$, $(\mu * \chi)(p^e) = \sum_{j=0}^e \mu(p^j) \chi(p)^{e-j} = \chi(p)^e - \chi(p)^{e-1}$ (with $0^0 = 1$), which is $\chi(p)^{e-1}(\chi(p) - 1)$, and $\mu * \chi$ is multiplicative. If $\chi(p) = 1$ the value is 0 for every $e \geq 1$. For χ_{-3} : $\chi_{-3}(3) = 0$ gives -1 at 3 and 0 at 3^e , $e \geq 2$; $\chi_{-3}(p) = -1$ gives $2(-1)^e$ at p^e . $\square$

So $G_{-3}(x) = 2a(x)$ is a signed count of the integers built from the primes $\equiv 2 \pmod{3}$ and at most one factor 3, each weighted $\pm 2^{\nu(n)}$; the primes $\equiv 1 \pmod{3}$ do not enter it, and neither do the multiples of 9. Up to $2 \cdot 10^5$ these integers are 28.59% of all integers. The same holds at every modulus: the unit classes of the tower modulo 3^k are built from $\mu * \chi$ for the characters $\chi \bmod 3^k$, which vanish on every multiple of a prime $\equiv 1 \pmod{3^k}$.

2.3. The Triad at a Zero and the Waves

Proof of Theorem 3. For $c = 1, 2$, $A_c(s) = \sum_{n \geq 0} (3n + c)^{-s} = 3^{-s} \zeta(s, c/3)$, and $A_0(s) = 3^{-s} \zeta(s)$; these continue to $\mathbb{C} \setminus \{1\}$ [1, Ch. 12]. The identities $A_0 + A_1 + A_2 = \zeta$ and $A_1 - A_2 = L(s, \chi_{-3})$ hold

for $\sigma > 1$ and hence everywhere, so $A_1 + A_2 = (1 - 3^{-s})\zeta(s)$, and at a zero ρ of ζ , $A_0(\rho) = 0$ and $A_1(\rho) = -A_2(\rho) = \frac{1}{2}L(\rho, \chi_{-3})$. Modulo 4, $B_0 = 4^{-s}\zeta(s)$, $B_2 = 4^{-s}\zeta(s, \frac{1}{2}) = 4^{-s}(2^s - 1)\zeta(s)$, $B_1 + B_3 = (1 - 2^{-s})\zeta(s)$ and $B_1 - B_3 = L(s, \chi_{-4})$. The Dirichlet series whose summatory function is $S_c^{(m)}$ is $\sum_{r \equiv c} r^{-s} \sum_n \mu(n) n^{-s} = m^{-s} \zeta(s, c/m) / \zeta(s)$, and at a simple zero the residue of $m^{-s} \zeta(s, c/m) x^s / (s \zeta(s))$ is $m^{-\rho} \zeta(\rho, c/m) x^\rho / (\rho \zeta'(\rho))$, which, when the zeros are simple, is the coefficient of x^ρ in the explicit formula, by the argument of [58, Theorem 14.27] for $M(x)$. $\square$

The residue vanishes for the class 0 mod 3 and the classes 0, 2 mod 4, which are constant. For a zero $\rho = \frac{1}{2} + i\gamma$ and its conjugate, the waves of $M(x)/\sqrt{x}$, $a(x)/\sqrt{x}$ and $b(x)/\sqrt{x}$ in the doubling variable u have pitch $\gamma \log 2/2\pi$ and amplitudes

$$\frac{2}{|\rho \zeta'(\rho)|}, \quad \frac{|L(\rho, \chi_{-3})|}{|\rho \zeta'(\rho)|}, \quad \frac{|L(\rho, \chi_{-4})|}{|\rho \zeta'(\rho)|}, \quad (6)$$

and the mean square over u of a sum of waves of distinct pitches is half the sum of their squared amplitudes:

$$e_M = \sum_{\gamma > 0} \frac{2}{|\rho \zeta'(\rho)|^2}, \quad e_a = \sum_{\gamma > 0} \frac{|L(\rho, \chi_{-3})|^2}{2|\rho \zeta'(\rho)|^2}, \quad e_b = \sum_{\gamma > 0} \frac{|L(\rho, \chi_{-4})|^2}{2|\rho \zeta'(\rho)|^2}. \quad (7)$$

2.4. Level Two in Eisenstein Integers

The group $(\mathbb{Z}/9\mathbb{Z})^\times$ is cyclic of order 6, generated by 2. Let $\omega = e^{2\pi i/3}$, so that $\omega^2 = -1 - \omega$, and let χ_9 be the cubic character with $\chi_9(2) = \omega$: $\chi_9(1) = \chi_9(8) = 1$, $\chi_9(2) = \chi_9(7) = \omega$, $\chi_9(4) = \chi_9(5) = \omega^2$. The six characters modulo 9 are $\chi_0, \chi_{-3}, \chi_9, \bar{\chi}_9, \chi_9 \chi_{-3}, \bar{\chi}_9 \chi_{-3}$, and all their values lie in the Eisenstein integers $\mathbb{Z}[\omega]$. Put $G_9 = G_{\chi_9}$ and $G_9^* = G_{\bar{\chi}_9 \chi_{-3}}$, which lie in $\mathbb{Z}[\omega]$, and $\text{Tr}(a_1 + a_2 \omega) = 2a_1 - a_2$ for integers a_1, a_2 . Since μ is real, $G_{\bar{\chi}_9} = \overline{G_9}$ and $G_{\bar{\chi}_9 \chi_{-3}} = \overline{G_9^*}$, and Theorem 1(ii) gives, for $x \geq 9$ and every unit v mod 9,

$$S_v^{(9)}(x) = \frac{1}{6} \left(\chi_{-3}(v) G_{-3}(x) + \text{Tr}(\bar{\chi}_9(v) G_9(x)) + \chi_{-3}(v) \text{Tr}(\bar{\chi}_9(v) G_9^*(x)) \right), \quad (8)$$

together with $S_0^{(9)}(x) = 1$ and $S_3^{(9)}(x) = -S_6^{(9)}(x) = \frac{1}{2} G_{-3}(x/3)$. The whole of level two is thus computed in exact integer arithmetic in $\mathbb{Z}[\omega]$, with no complex numbers.

2.5. The Tower at 10^{10}

The exact sieve to $X = 10^{10}$ gives $M(X) = -33,722$, $G_{-3}(X) = -25,880$, $G_{-4}(X) = -66,506$, $G_9(X) = 6679 + 32060\omega$, $G_9^*(X) = -34989 - 6254\omega$ and $G_{-3}(X/3) = 23,700$. By Theorem 1 and (8) these give the class values of Table 1. Independently, the leaves $M(X/r)$ give $\sum_{r \leq X} M(X/r) = 1$ and the same class values modulo 3, 4 and 9; for instance (8) at $v = 2$ reads $\frac{1}{6}(25880 + 57441 - 22481) = 10140$. For every integer $3 \leq x < 5000$ the splits of Theorem 1(i) hold exactly in their stated ranges.

Table 1. The rebuild law at $x = 10^{10}$ split by r mod 3, r mod 4 and r mod 9. Each row sums to 1; the classes 1, 4, 7 mod 9 sum to $S_1^{(3)}$ and the classes 2, 5, 8 to $S_2^{(3)}$.

class c	0	1	2	3	4	5	6	7	8
$S_c^{(3)}(10^{10})$	1	-12,940	12,940						
$S_c^{(4)}(10^{10})$	1	-33,253	0	33,253					
$S_c^{(9)}(10^{10})$	1	-18,051	10,140	11,850	-3,896	-9,017	-11,850	9,007	11,817

Table 2 gives the energy of every level up to $\kappa = 12$ and the largest class value; the leaves are reached at $\kappa = 21$, where $\mathcal{E}_\infty(10^{10}) = 20,349,625,023 = 2.0350 x$.

Table 2. Energy and largest class of the triad tower at $x = 10^{10}$, level by level. At every level the largest class is at most $0.3396\sqrt{x}$; from $\kappa = 3$ on it is the class of $r = 1$, whose value tends to $M(x) = -0.3372\sqrt{x}$ (it is $-33,724$ at $\kappa = 9$ and $-33,721$ at $\kappa = 12$).

κ	1	2	3	4	5	6	7
$\mathcal{E}_\kappa/x$	0.0335	0.1027	0.2713	0.2666	0.2932	0.3301	0.3534
$\max_c S_c /\sqrt{x}$	0.1294	0.1805	0.3331	0.3312	0.3396	0.3376	0.3375
κ	8	9	10	11	12	leaves	
$\mathcal{E}_\kappa/x$	0.3844	0.4111	0.5043	0.5189	0.5545	2.0350	
$\max_c S_c /\sqrt{x}$	0.3371	0.3372	0.3372	0.3372	0.3372	0.3372	

Waves

On the grid $u = 12 + 0.001j$, $2^{12} \leq x \leq 10^{10}$, the amplitudes of $M(x)/\sqrt{x}$, $a(x)/\sqrt{x}$ and $b(x)/\sqrt{x}$ at the pitches of the first twelve zeros agree with (6) to within 0.0003 (Table 3). The sixth zero has $|L(\rho_6, \chi_{-3})| = 0.0796$, and its wave is nearly silent in the triad charge: 0.0012 measured, 0.0011 predicted, against 0.0274 in M . The tenth zero has $|L(\rho_{10}, \chi_{-4})| = 0.1892$ and its wave is nearly silent in the four-split charge: 0.0028 measured, 0.0027 predicted. At the twelve zeros the computed residuals $|A_0(\rho)|$ and $|A_1(\rho) + A_2(\rho)|$ are of order 10^{-14} , the precision of the zeros.

Table 3. Wave amplitudes (units of $\sqrt{x}$) of M , of the triad charge a and of the four-split charge b at the pitches $\gamma \log 2/2\pi$ (turns per doubling) of the first twelve zeros: measured from the counts to 10^{10} and predicted by (6). The last two columns are $|A_1(\rho)|$ and $|B_1(\rho)|$.

γ	pitch	M meas.	pred.	a meas.	pred.	b meas.	pred.	$\frac{1}{2} L(\rho, \chi_{-3}) $	$\frac{1}{2} L(\rho, \chi_{-4}) $
14.1347	1.5593	.1784	.1783	.2584	.2582	.1988	.1987	1.4484	1.1143
21.0220	2.3191	.0836	.0837	.0585	.0587	.0469	.0468	.7011	.5598
25.0109	2.7591	.0583	.0583	.0523	.0525	.0541	.0541	.9000	.9286
30.4249	3.3564	.0505	.0504	.0254	.0253	.0552	.0551	.5013	1.0935
32.9351	3.6333	.0440	.0439	.0631	.0629	.0189	.0187	1.4321	.4262
37.5862	4.1464	.0274	.0275	.0012	.0011	.0359	.0358	.0398	1.3046
40.9187	4.5141	.0329	.0328	.0704	.0704	.0190	.0190	2.1484	.5810
43.3271	4.7797	.0253	.0252	.0140	.0139	.0454	.0453	.5540	1.8010
48.0052	5.2958	.0266	.0266	.0209	.0208	.0135	.0132	.7839	.4964
49.7738	5.4909	.0284	.0283	.0372	.0371	.0028	.0027	1.3118	.0946
52.9703	5.8436	.0154	.0156	.0083	.0084	.0050	.0050	.5374	.3192
56.4462	6.2270	.0150	.0150	.0139	.0139	.0094	.0093	.9269	.6236

Energies

The mean squares over the same grid, and over its two halves, agree with the sums (7) over the 2469 zeros with $0 < \gamma < 3000$, completed by a tail beyond 3000 estimated from the mean of γ^2 times the summand over $2000 < \gamma < 3000$ and the density $\frac{1}{2\pi} \log \frac{t}{2\pi}$ of the zeros (Table 4).

Table 4. Mean square over $2^{12} \leq x \leq 10^{10}$ in doublings, against the energies (7).

	measured	first half	second half	zeros < 3000	with tail
$M(x)/\sqrt{x}$	0.02900	0.02857	0.02943	0.02897	0.02904
$a(x)/\sqrt{x}$	0.04864	0.04869	0.04859	0.04868	0.04890
$b(x)/\sqrt{x}$	0.03033	0.03017	0.03048	0.03011	0.03022

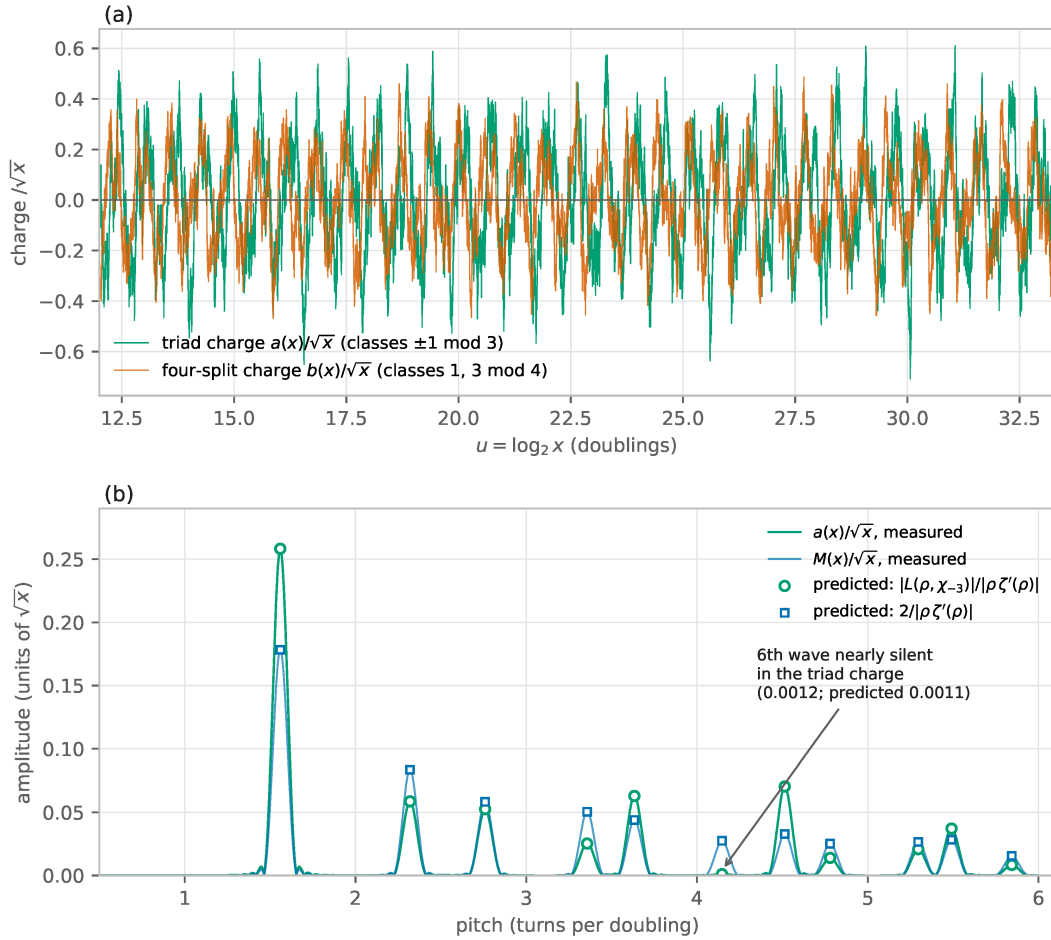


Figure 1. (a) The triad charge $a(x)/\sqrt{x}$ (classes $\pm 1 \pmod{3}$) and the four-split charge $b(x)/\sqrt{x}$ (classes $1, 3 \pmod{4}$) for $2^{12} \leq x \leq 10^{10}$, from the exact sieve. (b) Amplitude spectra in doublings of $a(x)/\sqrt{x}$ and $M(x)/\sqrt{x}$, with the amplitudes $|L(\rho, \chi_{-3})|/|\rho \zeta'(\rho)|$ (circles) and $2/|\rho \zeta'(\rho)|$ (squares) of (6) at the pitches of the zeros. The sixth wave is nearly silent in the triad charge.

3. Prime Gaps with the Prime Squares

Proof of Theorem 4. (i) Let $n \in [p, pp']$ be composite with no prime factor below p , and write $n = q_1 q_2 \cdots q_j$ with $j \geq 2$ and $p \leq q_1 \leq q_2 \leq \dots$. If $q_2 \geq p'$ then $n \geq pp'$, which is excluded; so $q_1 = q_2 = p$, the only prime in $[p, p']$. If $j \geq 3$ then $n \geq p^3 > pp'$, because $p^2 \geq 2p > p'$ by Bertrand's postulate. Hence $n = p^2$, which lies in the window since $p < p'$. Conversely every prime in $[p, pp']$ and p^2 have no prime factor below p . (ii) An odd square is $\equiv 1 \pmod{8}$, and a square prime to 3 is $\equiv 1 \pmod{3}$. (iii) For integers $\equiv \pm 1 \pmod{6}$, a gap $\equiv 2 \pmod{6}$ leads from the class -1 to the class $+1$, a gap $\equiv 4$ from $+1$ to -1 , and a gap $\equiv 0$ stays in its class. The crossings therefore alternate in direction, and the two counts differ by $[n_{\text{last}} \equiv 1] - [n_{\text{first}} \equiv 1] \in \{-1, 0, 1\}$. By (ii) every prime square ≥ 25 is $\equiv 1 \pmod{6}$. $\square$

Corollary 1 (each square opens the next wheel). *Let $W_p = \prod_{q < p} q$. On $[p, pp']$ the gaps between consecutive members of the primes together with p^2 are the gaps of the wheel of integers prime to W_p . The least composite member of this wheel is p^2 , and the next is pp' .*

For $p = 5$ the window is $[5, 35)$ and the wheel is that of 6: the numbers 5, 7, 11, 13, 17, 19, 23, 25, 29, 31 have gaps 2, 4, 2, 4, $\dots$. For $p = 7$ the window is $[7, 77)$ and the wheel is that of 30; for $p = 11$ it is $[11, 143)$ and the wheel of 210. We have checked the window law directly through $p = 19$. The wheel of 30 is symmetric about 30, since $\gcd(n, 30) = \gcd(60 - n, 30)$: its members in

(0, 60) lie at the distances 1, 7, 11, 13, 17, 19, 23, 29 from 30 on both sides, and $49 = 60 - 11$, $59 = 60 - 1$. With 49 inserted, the gaps from 7 to 67 are the gaps of the wheel of 30, namely 4, 2, 4, 2, 4, 6, 2, 6, twice.

3.1. Gaps and Classes Up to 61

From 5 to 59 the gaps of the primes are

$$2, 4, 2, 4, 2, 4, 6, 2, 6, 4, 2, 4, 6, 6,$$

and with 25 and 49 inserted they are

$$2, 4, 2, 4, 2, 4, 2, 4, 2, 6, 4, 2, 4, 2, 4, 6.$$

Table 5 lists the ten triads $6k - 1, 6k, 6k + 1$ up to 61. Without the squares the classes $6k - 1$ and $6k + 1$ hold 9 and 7 primes, and the four gaps of length 6 ($23 \rightarrow 29$, $31 \rightarrow 37$, $47 \rightarrow 53$, $53 \rightarrow 59$) pass over 25, 35, 49 and 55: three times over the class $+1$ and once over the class -1 . With the squares inserted the two classes stand $9 : 9$, each with exactly one composite not a square, $35 = 5 \cdot 7$ and $55 = 5 \cdot 11$, and the two remaining gaps of length 6 pass over these two, one in each class. The gaps $\equiv 2$ and $\equiv 4 \pmod{6}$ number 5 and 5 without the squares and 7 and 7 with them.

Table 5. The ten triads up to 61. The class $6k - 1$ holds nine primes and the composite 35; the class $6k + 1$ holds seven primes, the two squares 25 and 49, and the composite 55.

k	$6k - 1$	$6k$	$6k + 1$	k	$6k - 1$	$6k$	$6k + 1$
1	5	6	7	6	$35 = 5 \cdot 7$	36	37
2	11	12	13	7	41	42	43
3	17	18	19	8	47	48	$49 = 7^2$
4	23	24	$25 = 5^2$	9	53	54	$55 = 5 \cdot 11$
5	29	30	31	10	59	60	61

3.2. Beyond 61

Up to 10^8 the gaps between consecutive primes from 5 on number 1,616,470 ($\equiv 2 \pmod{6}$), 1,616,470 ($\equiv 4$) and 2,528,512 ($\equiv 0$). Table 6 compares the two classes with the prime squares counted at weight 0, 1 and $\frac{1}{2}$. At weight 1 the class $6k + 1$ pulls ahead as the squares accumulate; at weight $\frac{1}{2}$ the difference stays within the oscillation carried by the zeros of $L(s, \chi_{-3})$, of scale $\sqrt{x}/\log x$, which is 543 at 10^8 . Section 4 explains the weight $\frac{1}{2}$.

Table 6. The classes $6k - 1$ and $6k + 1$ of the primes up to x , the number of prime squares $p^2 \leq x$ with $p \geq 5$ (all in the class $6k + 1$), and $\pi(x; 6, 5) - \pi(x; 6, 1) - w \cdot \# \text{squares}$ for $w = 0, 1, \frac{1}{2}$.

x	$\pi(x; 6, 5)$	$\pi(x; 6, 1)$	squares	difference	squares at weight 1	squares at weight $\frac{1}{2}$
61	9	7	2	2	0	1
100	12	11	2	1	-1	0
10^3	86	80	9	6	-3	1.5
10^4	616	611	23	5	-18	-6.5
10^5	4,806	4,784	63	22	-41	-9.5
10^6	39,265	39,231	166	34	-132	-49
10^7	332,383	332,194	444	189	-255	-33
10^8	2,880,936	2,880,517	1,227	419	-808	-194.5

4. The Squares Carry the Lean

Proof of Theorem 5. (i) For $\sigma > 1$, $\sum_{(n,W)=1} n^{-s} = \sum_n n^{-s} \sum_{d|(n,W)} \mu(d) = \sum_{d|W} \mu(d) d^{-s} \zeta(s)$. The finite product $\prod_{q|W} (1 - q^{-s})$ is entire and has no zeros in $\sigma > 0$, because $|q^{-s}| < 1$ there. Likewise $\Phi_W(x) = \sum_{d|W} \mu(d) \lfloor x/d \rfloor = x \sum_{d|W} \mu(d)/d - \sum_{d|W} \mu(d) \{x/d\}$, with $\sum_{d|W} \mu(d)/d = \varphi(W)/W$; each $\{x/d\}$ has period $d \mid W$, and W has $2^{\nu(W)}$ divisors. (ii) $\log \zeta(s) = -\sum_p \log(1 - p^{-s}) =$

$\sum_p \sum_k p^{-ks}/k$, and J jumps by $1/k$ at each p^k . Then $\sum_k \mu(k)k^{-1}J(x^{1/k}) = \sum_{k,j} \mu(k)(kj)^{-1}\pi(x^{1/kj}) = \sum_n \pi(x^{1/n})n^{-1} \sum_{k|n} \mu(k) = \pi(x)$, all sums being finite. (iii) By definition $\pi(x) = J(x) - \frac{1}{2}\pi(\sqrt{x}) - \sum_{k \geq 3} \pi(x^{1/k})/k$, the last sum is $O(x^{1/3})$, and $\pi(\sqrt{x}) = \text{li}(\sqrt{x}) + O(\sqrt{x}e^{-c\sqrt{\log x}})$ by the prime number theorem [21, Ch. 18]. The formula for J is Riemann's [51], proved by von Mangoldt [62]. (iv) $\zeta(2s)/\zeta(s) = \zeta(2s) \cdot \zeta(s)^{-1}$ gives $\lambda(n) = \sum_{d^2|n} \mu(n/d^2)$, so $L(x) = \sum_{d^2 m \leq x} \mu(m) = \sum_{d \leq \sqrt{x}} M(x/d^2)$. (v) The left side is $\sum_{k \geq 2} \sum_{p^k \leq x} \log p ([p^k \equiv 1] - [p^k \equiv 5]) \pmod{6}$. Powers of 2 and 3 are not prime to 6. For $p \geq 5$, $p^2 \equiv 1 \pmod{6}$ by Theorem 4(ii), so $k = 2$ gives $\sum_{5 \leq p \leq \sqrt{x}} \log p = \theta(\sqrt{x}) - \log 6$, and the terms $k \geq 3$ total at most $\sum_{3 \leq k \leq \log_2 x} \theta(x^{1/k}) = O(x^{1/3} \log x)$. $\square$

Three Weights for One Square

Since $\sqrt{p^2} = p$, the prime square sits on its own square-root horizon, and three counts weigh it differently. The wheel of modulus W_p counts p^2 as a member, with weight 1 (Corollary 1); the primes leave it out, with weight 0; the logarithm of the Euler product counts it with weight $\frac{1}{2}$ (Theorem 5(ii)). By Theorem 5(i) every wheel series vanishes at every zero of ζ , yet its count carries no wave: its error is bounded and periodic, and for the wheel of $\prod_{p \leq 37} p$ the spectrum in doublings of the error divided by $\sqrt{x}$ has amplitude at most 0.0018 at every pitch in $[0.3, 6]$ up to 10^{10} (Section 12). The zeros become waves only through the logarithm, which turns each of them into a pole of ζ'/ζ , and the count that carries these waves with no term of order $\sqrt{x}/\log x$ is J , in which the squares have weight $\frac{1}{2}$. By Theorem 5(iii), $\pi(x) - \text{li}(x)$ contains in addition $-\frac{1}{2}\text{li}(\sqrt{x}) \sim -\sqrt{x}/\log x$: the lean of π against li is the square term. In the same way $L(x) - M(x) = \sum_{d \geq 2} M(x/d^2)$ carries the constant $1/\zeta(\frac{1}{2})$ of the pole of $\zeta(2s)$ at $s = \frac{1}{2}$ (Theorem 8(i)), and by Theorem 5(v) the difference $\theta(x; 6, 1) - \theta(x; 6, 5)$ carries $-\theta(\sqrt{x}) \sim -\sqrt{x}$, while $\psi(x; 6, 1) - \psi(x; 6, 5) = \psi(x, \chi_{-3}) + O(\log x)$ is, by its explicit formula [21, Ch. 19], a sum of waves from the zeros of $L(s, \chi_{-3})$. This is the mechanism of Chebyshev's bias identified by Rubinstein and Sarnak [53].

4.1. Measurements

The exact counts to 10^{10} (Section 12) give π , li , M , L , θ and ψ in the classes modulo 6 exactly. On the grid $u = 12 + 0.001j$, $2^{12} \leq x \leq 10^{10}$ (21,220 points), Table 7 compares four counts with the squares left out and with the squares at weight $\frac{1}{2}$.

Table 7. The lean of four prime counts over $2^{12} \leq x \leq 10^{10}$: mean over the grid, fraction of the grid where the count is positive, and number of sign changes on the grid. In ψ the prime power p^k has weight $\log p = \frac{1}{k} \log p^k$, so its squares enter with weight $\frac{1}{2}$ relative to their position; $\pi_{\square}(x) = \#\{p \geq 5 : p^2 \leq x\}$.

count	squares	mean	time above 0	sign changes
$(\pi(x) - \text{li}(x)) \log x / \sqrt{x}$	left out	-1.364	0.000	0
$(J(x) - \text{li}(x) + \log 2) \log x / \sqrt{x}$	weight $\frac{1}{2}$	+0.0013	0.501	1816
$L(x) / \sqrt{x} = \sum_d M(x/d^2) / \sqrt{x}$	d^2 -multiples in	-0.6834	0.000	0
$M(x) / \sqrt{x}$	d^2 -multiples out	-0.0046	0.484	826
$(\theta(x; 6, 1) - \theta(x; 6, 5)) / \sqrt{x}$	left out	-0.973	0.001	14
$(\psi(x; 6, 1) - \psi(x; 6, 5)) / \sqrt{x}$	weight $\frac{1}{2}$	-0.0039	0.502	1273
$(\pi(x; 6, 5) - \pi(x; 6, 1)) \log x / \sqrt{x}$	left out	+1.157	1.000	0
$(\pi(x; 6, 5) - \pi(x; 6, 1) - \frac{1}{2}\pi_{\square}(x)) \log x / \sqrt{x}$	weight $\frac{1}{2}$	+0.031	0.526	1408

The Balancing Weight

For $\pi_w(x) = \pi(x) + w\pi(\sqrt{x}) + \sum_{k \geq 3} \pi(x^{1/k})/k$, so that π_0 omits the squares and $\pi_{1/2} = J$, the mean of $(\pi_w(x) - \text{li}(x) + \log 2) \log x / \sqrt{x}$ over the grid is linear in w : it is -1.147 at $w = 0$ and +1.150 at $w = 1$, and it vanishes at $w^* = 0.4994$, with 0.4992 and 0.4997 on the two halves of the range. For the classes, the weight w that centres $\pi(x; 6, 5) - \pi(x; 6, 1) - w\pi_{\square}(x)$ is 0.514 (0.518 and 0.509 on the two halves).

The Waves are Unchanged

The amplitudes of $(J(x) - \text{li}(x)) \log x / \sqrt{x}$ at the first eight zeros are 0.1419, 0.0954, 0.0798, 0.0658, 0.0609, 0.0532, 0.0492 and 0.0462, against $2/|\rho| = 0.1414, 0.0951, 0.0799, 0.0657, 0.0607, 0.0532, 0.0489$ and 0.0462, and $\pi - \text{li}$ carries the same waves. The square weight moves the centre of the count and nothing else.

The Liouville Lean

$L(x) - M(x) = \sum_{d \geq 2} M(x/d^2)$ has mean $-0.6788\sqrt{x}$ over the grid, against $1/\zeta(\frac{1}{2}) = -0.6848$; it holds almost the whole lean of L , while M itself is centred. Tanaka [57] found $L(x) \leq 0$ for $2 \leq x < 906,150,257$, answering a question of Pólya [50]; in Table 7 the same count with its square multiples removed is M , which changes sign 826 times on the grid. Likewise $\pi(x) < \text{li}(x)$ at every grid point, although $\pi(x) - \text{li}(x)$ changes sign infinitely often [35], while $J(x) - \text{li}(x) + \log 2$ changes sign 1816 times.

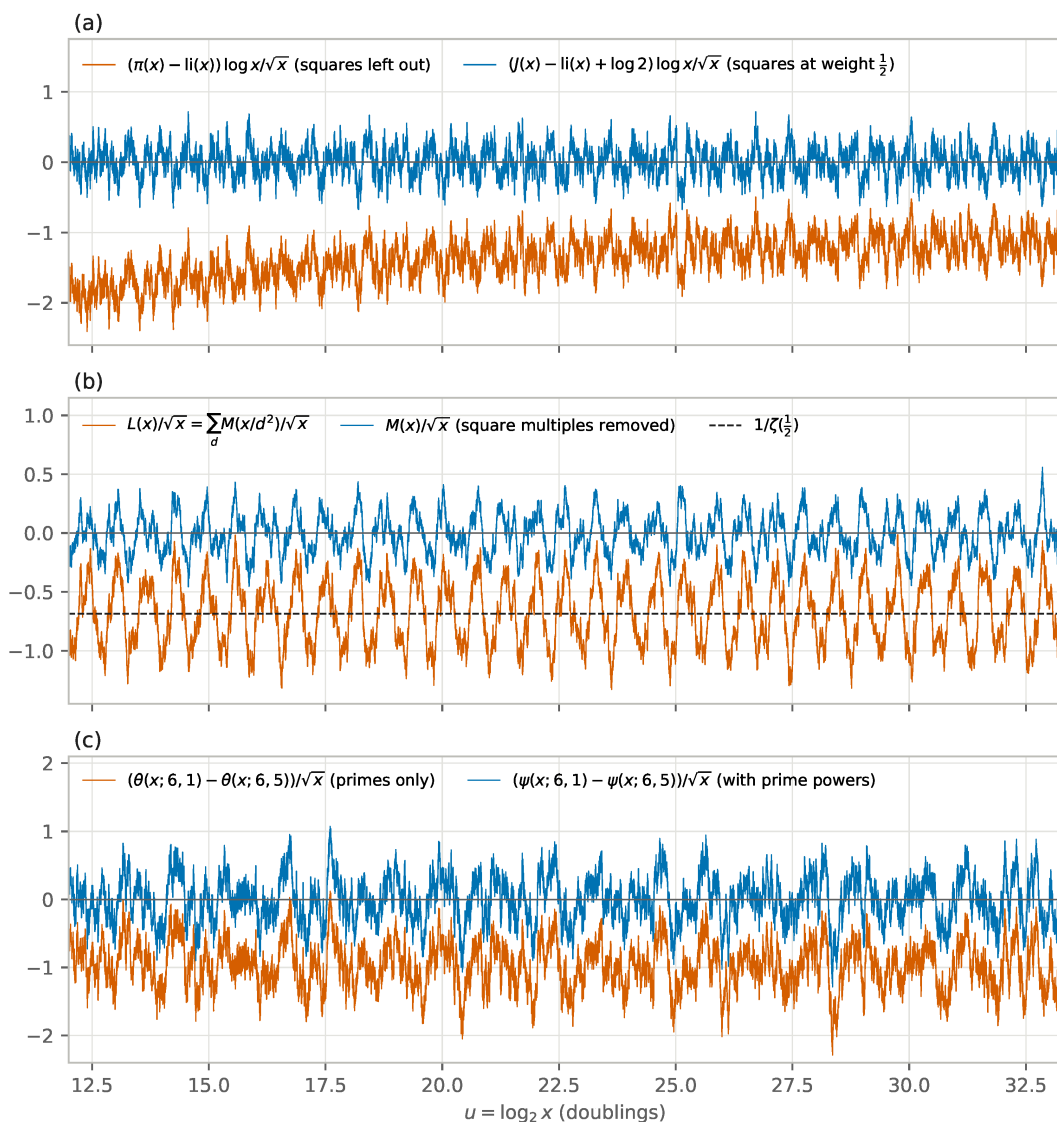


Figure 2. The squares carry the lean, $2^{12} \leq x \leq 10^{10}$. (a) $(\pi(x) - \text{li}(x)) \log x / \sqrt{x}$ and $(J(x) - \text{li}(x) + \log 2) \log x / \sqrt{x}$. (b) $L(x)/\sqrt{x}$ and $M(x)/\sqrt{x}$, with $1/\zeta(\frac{1}{2})$ dashed. (c) $(\theta(x; 6, 1) - \theta(x; 6, 5))/\sqrt{x}$ and $(\psi(x; 6, 1) - \psi(x; 6, 5))/\sqrt{x}$. In each panel the count with the squares at weight $\frac{1}{2}$ is centred and carries the same waves.

5. The Square-Root Horizon

We call $\sigma = \frac{1}{2}$ the square-root horizon: it is the line where the squares series $\sum_p p^{-2\sigma}$, and with it the part of $\log \zeta$ carried by the prime squares and higher powers, stops converging. Write $P(s) = \sum_p p^{-s}$ for $\sigma > 1$, so that $\log \zeta = P + S$.

Proof of Theorem 6. (i) For $\sigma > \frac{1}{2}$, $\sum_p \sum_{k \geq 2} |p^{-ks}|/k \leq \sum_p p^{-2\sigma} (1 - p^{-\sigma})^{-1} < \infty$, uniformly on compact sets, so S is analytic and $E = e^S$ is analytic and zero-free there. For $\sigma > 1$, $\log \zeta(s) = \sum_{m \geq 1} P(ms)/m$, and Möbius inversion, justified by absolute convergence, gives

$$P(s) = \sum_{k \geq 1} \frac{\mu(k)}{k} \log \zeta(ks), \quad S(s) = - \sum_{k \geq 2} \frac{\mu(k)}{k} \log \zeta(ks). \quad (9)$$

For $k \geq 2$ and $\sigma > \frac{1}{2}$, $|\log \zeta(ks)| \leq \log \zeta(k\sigma) \leq \zeta(k\sigma) - 1 \ll 2^{-k\sigma}$, so the second series converges to an analytic function of $\sigma > \frac{1}{2}$, which is S there. Since E has neither zeros nor poles in $\sigma > \frac{1}{2}$, $F = \zeta/E$ is meromorphic there with the zeros of ζ and their orders, and $F = e^P$ for $\sigma > 1$. (ii) $\log |E(\sigma + it)| = \operatorname{Re} S = \sum_p \sum_{k \geq 2} p^{-k\sigma} k^{-1} \cos(kt \log p)$, so, with $\theta = kt \log p$,

$$3 \log E(\sigma) + 4 \log |E(\sigma + it)| + \log |E(\sigma + 2it)| = \sum_p \sum_{k \geq 2} \frac{p^{-k\sigma}}{k} (3 + 4 \cos \theta + \cos 2\theta),$$

an absolutely convergent series of terms $\frac{p^{-k\sigma}}{k} \cdot 2(1 + \cos \theta)^2 \geq 0$. (iii) Since $\mu(2) = -1$, (9) gives $2S(s) = \log \zeta(2s) - 2R(s)$ with $R(s) = \sum_{k \geq 3} \mu(k) k^{-1} \log \zeta(ks)$. For $k \geq 3$ and $\sigma > \frac{1}{3}$, $\operatorname{Re}(ks) > 1$ and the bound of (i) makes R analytic in $\sigma > \frac{1}{3}$. Hence $E(s)^2 = \zeta(2s)e^{-2R(s)}$ continues meromorphically to $\sigma > \frac{1}{3}$. The factor e^{-2R} has no zeros or poles; $\zeta(2s)$ has its only pole at $s = \frac{1}{2}$ and no zeros with $\operatorname{Re}(2s) \geq 1$ [58, §3.3]. Since $\zeta(\frac{1}{2}) = -1.4603 \dots \neq 0$, $F(s)^2 = \zeta(s)^2 e^{2R(s)} / \zeta(2s)$ has a simple zero at $s = \frac{1}{2}$. For real $\sigma \rightarrow \frac{1}{2}+$, $\zeta(2\sigma) = (2\sigma - 1)^{-1} + \gamma_E + O(2\sigma - 1)$ [58, §2.1], so $\log E(\sigma) + \frac{1}{2} \log(2\sigma - 1) = -R(\sigma) + O(2\sigma - 1) \rightarrow -R(\frac{1}{2}) = C_E$, and $\log |F(\sigma)| - \frac{1}{2} \log(2\sigma - 1) = \log |\zeta(\sigma)| - (\log E(\sigma) + \frac{1}{2} \log(2\sigma - 1)) \rightarrow \log |\zeta(\frac{1}{2})| - C_E$. (iv) Let $\rho = \frac{1}{2} + i\gamma$ have multiplicity m . As $\sigma \rightarrow \frac{1}{2}+$, $3 \log |\zeta(\sigma)| \rightarrow 3 \log |\zeta(\frac{1}{2})|$, $\log |\zeta(\sigma + 2i\gamma)|$ tends to $\log |\zeta(\frac{1}{2} + 2i\gamma)|$ or to $-\infty$, and $4 \log |\zeta(\sigma + i\gamma)| = 4m \log(\sigma - \frac{1}{2}) + O(1) \rightarrow -\infty$. So $Q_\zeta(\sigma, \gamma) \rightarrow -\infty$. By (iii), $3 \log E(\sigma) = \frac{3}{2} \log \frac{1}{2\sigma - 1} + O(1)$, while $\log |E(\sigma + i\gamma)|$ and $\log |E(\sigma + 2i\gamma)|$ stay bounded, because E^2 is analytic and zero-free at $\frac{1}{2} + i\gamma$ and $\frac{1}{2} + 2i\gamma$; hence $Q_F = Q_\zeta - Q_E \rightarrow -\infty$ as well. $\square$

Remark 1 (the half-order zero is the square lean). The expansion $P(s) = \log \zeta(s) - \frac{1}{2} \log \zeta(2s) - \frac{1}{3} \log \zeta(3s) - \frac{1}{5} \log \zeta(5s) + \frac{1}{6} \log \zeta(6s) - \dots$ of (9) is, term by term, Riemann's expansion $\pi(x) = J(x) - \frac{1}{2} J(\sqrt{x}) - \frac{1}{3} J(x^{1/3}) - \dots$ of Theorem 5(ii). The term $-\frac{1}{2} \log \zeta(2s)$, which near $s = \frac{1}{2}$ is $\frac{1}{2} \log(2s - 1) + O(1)$, is the half-order zero of $F = e^P$ at the horizon, and in the count it is $-\frac{1}{2} J(\sqrt{x}) = -\frac{1}{2} \operatorname{li}(\sqrt{x}) + \dots$, the lean of π against li measured in Section 4. The half-order pole of E at $s = \frac{1}{2}$ is the same square term seen from the squares part.

Remark 2 (edge and horizon). For ζ itself the squared pair is nonnegative for $\sigma > 1$, $\zeta(\sigma)^3 |\zeta(\sigma + it)|^4 |\zeta(\sigma + 2it)| \geq 1$ [37], [58, §3.3], from the identity $3 + 4 \cos \theta + \cos 2\theta = 2(1 + \cos \theta)^2 \geq 0$ [58, (3.3.1)], which gives $\zeta(1 + it) \neq 0$ and, by the functional equation, $\zeta(it) \neq 0$. The inequality for ζ splits as $Q_\zeta = Q_E + Q_F$: the prime part $Q_F = \sum_p p^{-\sigma} \cdot 2(1 + \cos(t \log p))^2 \geq 0$ holds for $\sigma > 1$, and the squares part $Q_E \geq 0$ holds all the way down to the horizon by Theorem 6(ii). The zeros of ζ in $\sigma > \frac{1}{2}$ are exactly those of the prime factor F (Theorem 6(i)). At the horizon the squares part has a half-order pole and the prime factor a half-order zero, and by Theorem 6(iv) the squared pair of ζ and of F is negative arbitrarily close to the horizon at the height of every zero on it.

5.1. Measurements

The Squares Part

By (9), $\log |E(s)| = -\sum_{k \geq 2} \mu(k) k^{-1} \log |\zeta(ks)|$ involves ζ only in $\sigma > 1$. At 200 random points with $\frac{1}{2} < \sigma < 1$ and $1 < t < 61$ the squared pair Q_E has minimum 0.3355. Table 8 shows the half-order limits of Theorem 6(iii).

Table 8. The half-order pole of the squares part E and the half-order zero of the prime factor $F = \zeta/E$ at $s = \frac{1}{2}$. The two limits add up to $\log |\zeta(\frac{1}{2})| = 0.37868$.

σ	0.6	0.55	0.51	0.505	0.501	0.5001	limit
$\log E(\sigma) + \frac{1}{2} \log(2\sigma - 1)$	0.29424	0.31979	0.35315	0.35842	0.36285	0.36387	0.36399
$\log F(\sigma) - \frac{1}{2} \log(2\sigma - 1)$	0.37495	0.19824	0.05258	0.03374	0.01852	0.01507	0.01469

The Reach of the Squared Pair of ζ

For a zero $\frac{1}{2} + i\gamma$ let its reach be the largest $\sigma \in [0.5005, 1]$ at which $\min_{|t-\gamma| \leq 0.3} Q_\zeta(\sigma, t) < 0$. At the first 41 zeros and every 25th zero after them, 138 zeros in all up to height 3000, the reach lies between 0.546 and 0.847 (Table 9, Figure 3). For the prime factor alone the reach at the first twelve zeros is 0.888, 0.862, 0.855, 0.869, 0.848, 0.798, 0.837, 0.806, 0.852, 0.856, 0.775 and 0.773. Above the reach the squared pair of ζ is nonnegative at the sampled points of the window, and at such points ζ has no zero; below it the band of negative values extends down to the zero on the horizon, as Theorem 6(iv) requires.

Table 9. Reach of the negative band of the squared pair Q_ζ beside the zeros of ζ .

height of the zero	< 50	50–100	100–300	300–1000	1000–2000	2000–3000
zeros	10	19	15	21	35	38
mean reach	0.802	0.761	0.747	0.727	0.711	0.693
largest reach	0.839	0.818	0.837	0.847	0.845	0.825

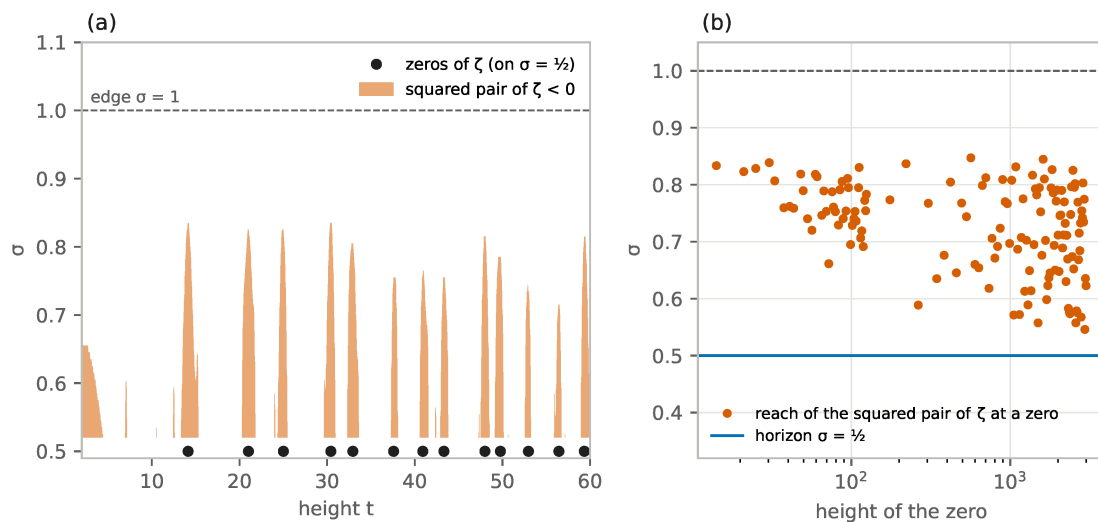


Figure 3. (a) The region $\frac{1}{2} < \sigma \leq 1.1$, $2 \leq t \leq 60$ where the squared pair Q_ζ is negative, and the zeros of ζ on $\sigma = \frac{1}{2}$. (b) The reach of the negative band at 138 zeros up to height 3000. The squares part satisfies $Q_E \geq 0$ everywhere above the horizon.

6. The Prime Term at Every Height

By Theorem 6 every zero of ζ in $\sigma > \frac{1}{2}$ is a zero of the prime factor F , whose logarithm is the prime term $P(s) = \sum_p p^{-s}$ for $\sigma > 1$. At height t the prime term is the sum of the unit vectors $p^{-it} = e^{-it \log p}$

weighted by $p^{-\sigma}$, and its phases $t \log p$ behave as independent in the sense of Theorem 7(i): any finitely many of them are jointly equidistributed.

Proof of Theorem 7(i)–(iii). (i) By unique factorisation the numbers $\log p$ are linearly independent over $\mathbb{Q}$: an integer relation $\sum a_p \log p = 0$ gives $\prod p^{a_p} = 1$, so every $a_p = 0$. Hence for every nonzero integer vector $(k_p)_{p \leq y}$, $\sum k_p \log p \neq 0$ and $\frac{1}{T} \int_0^T e^{it \sum_p k_p \log p} dt \rightarrow 0$, and Weyl's criterion [61] gives the equidistribution of $t(\log p/2\pi)_{p \leq y}$ modulo 1; this is Kronecker's theorem in Weyl's form [31, Ch. XXIII]. (ii) $\mathbb{E} e^{i(\theta_p - \theta_q)} = [p = q]$ gives the variance. For $\sigma > \frac{1}{2}$ the independent terms have mean 0 and summable variances, so the series converges almost surely [32], [23, Theorem 2.5.6]. For $\sigma \leq \frac{1}{2}$ the real parts $p^{-\sigma} \cos \theta_p$ are independent, have mean 0, are bounded by 1 and have variances $\frac{1}{2} p^{-2\sigma}$ with divergent sum, so by Kolmogorov's three-series theorem with truncation level 1 [23, Theorem 2.5.8] their sum does not converge almost surely; by Kolmogorov's zero-one law it diverges almost surely. (iii) Let $Z_y = \prod_{p \leq y} (1 - p^{-\sigma} e^{i\theta_p})^{-1} = \prod_{p \leq y} \sum_{j \geq 0} p^{-j\sigma} e^{ij\theta_p}$. Each factor has mean 1 and mean square $\sum_j p^{-2j\sigma} = (1 - p^{-2\sigma})^{-1}$, so by independence $\mathbb{E}|Z_y|^2 = \prod_{p \leq y} (1 - p^{-2\sigma})^{-1}$. For $y < y'$ let $Z_{y,y'} = Z_{y'}/Z_y$, which is independent of Z_y and has mean 1; then $\mathbb{E}|Z_{y'} - Z_y|^2 = \mathbb{E}|Z_y|^2 (\mathbb{E}|Z_{y,y'}|^2 - 1) \leq \zeta(2\sigma) (\prod_{y < p \leq y'} (1 - p^{-2\sigma})^{-1} - 1) \rightarrow 0$. So Z_y converges in mean square, and $\mathbb{E}|Z_\sigma|^2 = \lim \mathbb{E}|Z_y|^2 = \zeta(2\sigma)$. Also $\log Z_y = \sum_{p \leq y} (p^{-\sigma} e^{i\theta_p} + O(p^{-2\sigma}))$ converges almost surely by (ii), so Z_y converges almost surely to $\exp(\lim \log Z_y) \neq 0$, which is Z_σ . $\square$

Part (iv) consists of classical theorems: the mean square is Theorem 7.2 of [58], due to Hardy and Littlewood; $N(\sigma, T) = O(T)$ is Theorem 9.15(A) of [58], due to Bohr and Landau [12]; the bound $N(\sigma, T) = O(T^{1-(\sigma-\frac{1}{2})/4} \log T)$ is Theorem 9.19(C) of [58], due to Selberg [54]; and the limiting distribution is due to Bohr and Jessen [10,11]. Since $N(T) \sim \frac{T}{2\pi} \log T$, the zeros with $\beta > \sigma$ are a proportion $O(1/\log T)$ of the zeros up to height T , for every fixed $\sigma > \frac{1}{2}$.

The squares series appears twice. In the model, the variance of the prime term is $\sum_p p^{-2\sigma}$ and the mean square of the Euler product is $\zeta(2\sigma) = \sum_n n^{-2\sigma}$ (Theorem 7(ii),(iii)); for ζ itself, the mean square over heights is the same $\zeta(2\sigma)$ (Theorem 7(iv)). Both are finite exactly for $\sigma > \frac{1}{2}$. For $\frac{1}{2} < \sigma < 1$ the mean square has the second main term

$$\int_0^T |\zeta(\sigma + it)|^2 dt = \zeta(2\sigma)T + \frac{(2\pi)^{2\sigma-1} \zeta(2-2\sigma)}{2-2\sigma} T^{2-2\sigma} + \Delta_\sigma(T), \quad (10)$$

where Matsumoto [36] proves $\Delta_\sigma(T) = O(T^{1/(1+4\sigma)} \log^2 T)$ for $\frac{1}{2} < \sigma < \frac{3}{4}$.

6.1. Measurements

At 6000 heights drawn uniformly from [5000, 6000] we computed $\zeta(\sigma + it)$ for $\sigma = 0.6, 0.7, 0.8$. The model value is $\log |Z_\sigma|$ with the primes up to 10^6 and independent uniform phases, plus a Gaussian term with the variance $\frac{1}{2} \sum_{p > y} p^{-2\sigma} \approx y^{1-2\sigma} / (2(2\sigma-1) \log y)$, $y = 10^6$, of the omitted primes; 6000 model values were drawn for each σ . Table 10 and Figure 4 compare the two.

Table 10. The value distribution of $\zeta(\sigma + it)$ at 6000 random heights in [5000, 6000] against the independent-phase model. The fourth column is the mean of $|\zeta|^2$ over [5000, 6000] given by the two main terms of (10).

σ	mean $ \zeta ^2$	$\zeta(2\sigma)$	by (10)	sd of $\log \zeta $	model sd	0.1% quantile	model
0.6	4.40	5.59	4.45	0.882	0.888	-2.374	-2.444
0.7	3.00	3.11	2.98	0.737	0.719	-1.836	-1.746
0.8	2.18	2.29	2.27	0.611	0.623	-1.456	-1.515

The Deepest Dips

On the grid $t = 10 + 0.02j$, $10 \leq t < 3000$, the least value of $\log |\zeta(0.6 + it)|$ is -3.007 ($|\zeta| = 0.049$), at $t = 1329.12$, between the close zeros 1329.044 and 1329.205; the model's quantile at $1/2469$, one value per zero in the range, is -2.585 and the least of 50,000 model values, with the primes up to 10^5 ,

is -2.941 . At $\sigma = 0.7$ the least value is -2.214 , at the same height, against -1.911 and -2.203 . The deepest dips off the line sit beside zeros on the line, and at every computed point $|\zeta(\sigma + it)|$ is positive.

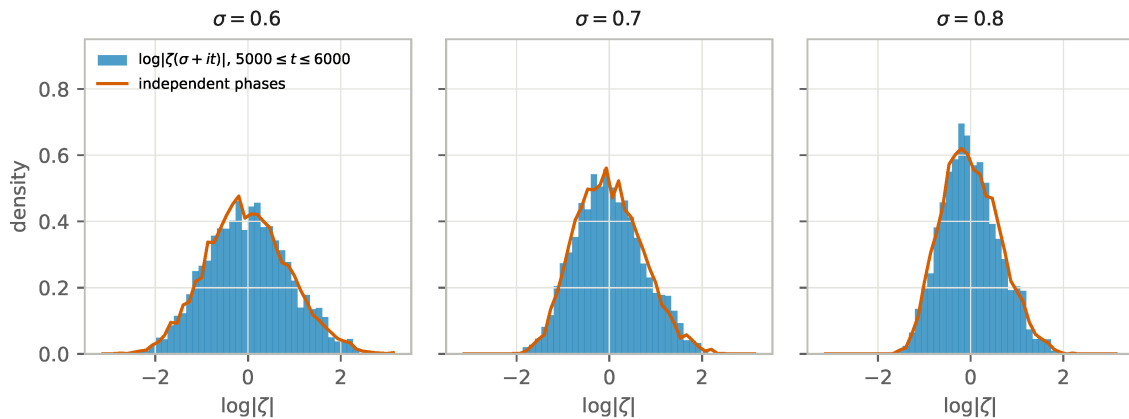


Figure 4. Distribution of $\log |\zeta(\sigma + it)|$ at 6000 random heights $5000 \leq t \leq 6000$ for $\sigma = 0.6, 0.7, 0.8$ (histograms), and of the independent-phase model $\log |Z_\sigma|$ (curves).

7. The Axiom of Harmonic Independence

Axiom 1 (harmonic independence). *Harmonic independence is the absence of an integer relation. The prime waves are related only through a mechanism present in their construction. Because Unique Factorization guarantees that no integer relation exists between the primes, their wave frequencies are strictly linearly independent. A harmonic resonance between the primes beyond this explicit construction—such as a zero off the critical line—would be a relation without a mechanism.*

The Waves and Their Frequencies

At height t the prime p enters the prime term $P(s) = \sum_p p^{-s}$ as the wave $p^{-it} = e^{-it \log p}$, weighted by $p^{-\sigma}$, of frequency $\log p / 2\pi$ in t ; measured in doublings of p it is the clock of Section 11. An integer relation between the frequencies is a vector (k_p) of integers, not all zero and finitely many nonzero, with $\sum_p k_p \log p = 0$, that is $\prod_p p^{k_p} = 1$. Unique factorisation excludes every such vector. This is the first step of the proof of Theorem 7(i), and through Kronecker's theorem in Weyl's form it makes the phases $t \log p$ of any finitely many primes jointly equidistributed on their torus: no finite set of prime waves holds a fixed phase relation over the heights, so the first part of the axiom is a theorem.

The Mechanism in the Construction

The waves are joined in ζ by one mechanism, the Euler product $\zeta(s) = \prod_p (1 - p^{-s})^{-1}$, whose expansion into $\sum_n n^{-s}$ is unique factorisation itself. The squares part E and the prime factor F of Theorem 6 are this mechanism divided into its terms of order two and higher and its prime term. The relations among the waves measured in this paper are made by it: the lean carried by the squares at weight $\frac{1}{2}$ (Theorem 5), the variance $\sum_p p^{-2\sigma}$ of the prime term (Theorem 7(ii)), and the half-order zero of F at $s = \frac{1}{2}$ that matches the half-order pole of E (Theorem 6).

A Zero Off the Line

By Theorem 6(i) a zero $\rho = \beta + i\gamma$ with $\beta > \frac{1}{2}$ is a zero of the prime factor F . At the height γ it would be a relation among the prime waves weighted by $p^{-\beta}$, a resonance beyond the independence of their frequencies and beyond the relations that the Euler product supplies. The model in which the waves carry no relation at all, the independent phases of Theorem 7, has this form: there the random Euler product Z_σ converges and is nonzero almost surely for every $\sigma > \frac{1}{2}$ (Theorem 7(iii)). The axiom states as a principle that ζ admits no relation among its prime waves beyond its construction; the theorems above give its first part, the linear independence of the frequencies, and with the axiom every zero of ζ in the critical strip lies on the line $\sigma = \frac{1}{2}$.

8. The Forward and Reversed Terms at the Midpoint

The approximate functional equation of Hardy and Littlewood [27,28], in the form of Riemann and Siegel [58, §§4.12–4.16], [56], writes ζ as a forward pattern, its reversal and a remainder:

$$\zeta(s) = \sum_{n \leq N} n^{-s} + \chi(s) \sum_{n \leq N} n^{s-1} + R_N(s), \quad \chi(s) = 2^s \pi^{s-1} \sin \frac{\pi s}{2} \Gamma(1-s), \quad N = \left\lfloor \sqrt{t/2\pi} \right\rfloor. \quad (11)$$

Theorem 11 (termwise balance). *Let $s = \sigma + it$ with $t > 0$. For every $n \geq 1$, $|\chi(s)n^{s-1}|/|n^{-s}| = |\chi(s)|n^{2\sigma-1}$. This ratio is independent of n if and only if $\sigma = \frac{1}{2}$, where it equals 1. For $\sigma \neq \frac{1}{2}$ it equals 1 for at most one real n , which is $\sqrt{t/2\pi}(1 + O_\sigma(1/t))$.*

Proof. $|n^{s-1}|/|n^{-s}| = n^{2\sigma-1}$. The functional equation gives $\chi(s)\chi(1-s) = 1$ and $\overline{\chi(s)} = \chi(\bar{s})$; on $\sigma = \frac{1}{2}$, $1-s = \bar{s}$, so $|\chi(s)|^2 = 1$. For $\sigma \neq \frac{1}{2}$, $n^{2\sigma-1}$ is strictly monotone in n , and $|\chi(s)| = (t/2\pi)^{1/2-\sigma}(1 + O_\sigma(1/t))$ by Stirling's formula [58, §4.12]. $\square$

The Axiom at the Midpoint

On the critical line every forward term is balanced against its reversed partner, and off it no two pairs of terms are balanced at once. A zero of ζ is a point at which the completed forward pattern and its reversal cancel. On the line the cancellation is between balanced partners; off it, it would require a relation among unbalanced terms that the construction (11) does not supply, which is what Axiom 1 excludes. The same structure is exact in a small graph, the medial antiprism: a stack of six triangles, each turned through 180° against the next, in which every vertex is joined to every side midpoint of the neighbouring layer. Its lines form two strands, exchanged by the reversal of the vertical order. If the midpoint nodes are moved to the fraction f along their sides, lines of the two strands meet in each gap exactly three times at height f and three times at height $1-f$, and only at the midpoint placement $f = \frac{1}{2}$ do all meetings lie on the middle plane.

A Mixed Construction

The Davenport–Heilbronn function $f(s) = \frac{1-i\zeta}{2}L(s, \chi) + \frac{1+i\zeta}{2}L(s, \bar{\chi})$, with χ the character modulo 5 with $\chi(2) = i$ and $\zeta = (\sqrt{10-2\sqrt{5}}-2)/(\sqrt{5}-1)$, satisfies a functional equation of the same kind as that of ζ and has zeros off the critical line [20], [58, §10.25]. It is the case that the axiom predicts. Its two Euler products are added with the relative weight $(1+i\zeta)/(1-i\zeta)$, which equals the root number $\varepsilon(\chi) = \tau(\chi)/(i\sqrt{5})$; both have argument 0.5535744, so the mixing is forced by the functional equation. The logarithm of ζ lives on the prime powers and gives every prime square the weight $\frac{1}{2}$. The logarithm of f has nonzero coefficients at 130 of the 302 integers up to 400 that are not prime powers, beginning with $b_6 = 1.080701$, and its square weights b_{p^2}/b_p^2 are $\frac{1}{2}$ where $\chi(p)$ is real and -12.891435 where $\chi(p) = \pm i$. Its zero $0.808517182 + 85.699348485i$ lies off the line. The zeros of f off the line are relations between prime waves that its construction supplies, which is what the axiom allows.

8.1. The Pair Without Its Remainder

The truncated pair $Z_N(s) = D_N(s) + \chi(s)D_N(1-s)$, $D_N(s) = \sum_{n \leq N} n^{-s}$, satisfies $Z_N(s) = \chi(s)Z_N(1-s)$, the functional equation of ζ , so its zeros off the line come in pairs $\sigma, 1-\sigma$. Unlike ζ , it has many. Scanning each range of constant N for $1 \leq N \leq 30$ (heights $6.3 \leq t \leq 6038$), we located 116 zeros of Z_N with $\frac{1}{2} < \sigma \leq 1.098$, each refined by Newton's method to $|Z_N| < 10^{-9}$ (Table 11, Figure 5). At every one of them the complete ζ is far from zero: $0.165 \leq |\zeta(s)| \leq 1.161$. The argument principle on a box of width 0.16 and height 0.8 about each of the 24 lowest finds no zero of ζ off the line, and all zeros of ζ in this range lie on the line [46]. At these heights the prime clocks are anti-aligned: the sum $\sum_{p \leq 400} \cos(t \log p)/\sqrt{p}$ lies at median percentile 16.7 among random heights nearby, and 33 of the 116 lie below the tenth percentile. The remainder R_N closes every one of these meetings back onto the line.

Table 11. Zeros of the truncated pair Z_N with $\sigma > \frac{1}{2}$ located below height 6000, and the complete ζ at them. Clock percentile: of $\sum_{p \leq 400} \cos(t \log p) / \sqrt{p}$ among 2000 random heights within 200 of t .

heights	zeros located	largest σ	least $ \zeta $ there	median clock percentile
[0, 1000)	20	1.098	0.290	10.2
[1000, 2000)	27	0.968	0.255	17.1
[2000, 3000)	12	1.051	0.208	17.9
[3000, 4000)	19	0.986	0.165	29.6
[4000, 5000)	21	0.980	0.254	17.2
[5000, 6000)	17	0.860	0.185	18.9

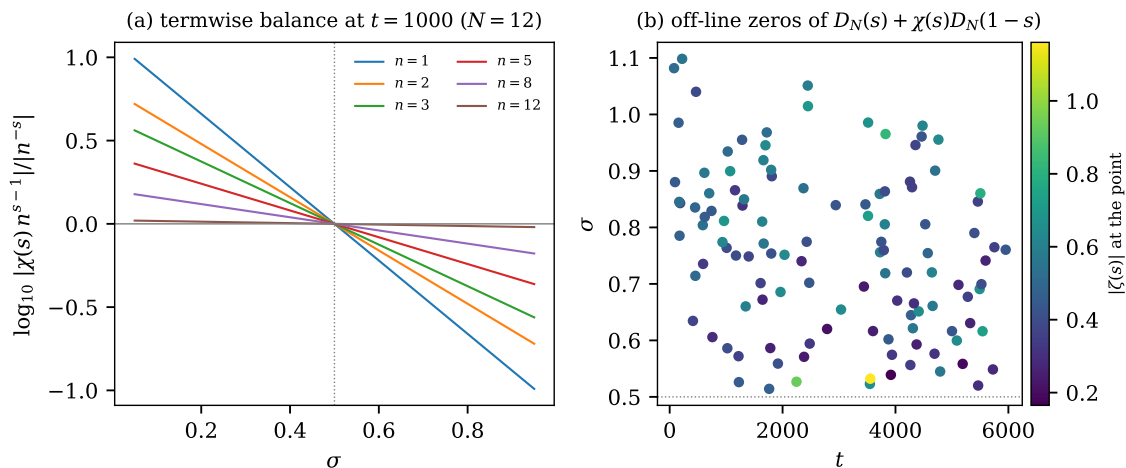


Figure 5. (a) The ratio $|\chi(s)n^{s-1}|/|n^{-s}|$ at $t = 1000$ for several n : every pair of terms is balanced exactly at $\sigma = \frac{1}{2}$ (Theorem 11). (b) The 116 located zeros of the truncated pair with $\sigma > \frac{1}{2}$, coloured by $|\zeta|$ at the same point.

8.2. Positive Chains

Following [24], define $\gamma(n)$ by $8\zeta(\frac{1}{2} + z) = \sum_{n \geq 0} \gamma(n) z^{2n}/n!$, where $\zeta(s) = \frac{1}{2}s(s-1)\pi^{-s/2}\Gamma(\frac{s}{2})\zeta(s)$, and the Jensen polynomials $J^{d,n}(X) = \sum_{j=0}^d \binom{d}{j} \gamma(n+j) X^j$. Pólya proved that RH holds if and only if every $J^{d,n}$ is hyperbolic, that is, has only real roots [47]; Griffin, Ono, Rolin and Zagier proved hyperbolicity for every d and all sufficiently large n , and for $d \leq 8$ and all n [24, Theorems 1, 2], the latter with a proof resting on a numerical check for $n \leq 10^6$; an effective form of the first theorem is in [25]. For $n = 0$, $J^{d,0}$ is known to be hyperbolic for $d \leq 2 \cdot 10^{17}$ [16]. A real polynomial p has only real zeros if and only if the J -fraction of p'/p , produced by the Euclidean algorithm on p and p' , has positive couplings, and its zeros are moreover distinct if and only if the fraction has full length; then p is the characteristic polynomial of a Jacobi matrix with those positive off-diagonal entries [60, Theorem 43.1], [17, Ch. I]. RH is therefore the statement that every Jensen polynomial of Ξ is a chain with positive couplings, the chain ending early only at a repeated real zero. Positive coupling keeps the zeros real.

We computed $\gamma(n)$ for $n \leq 820$ to 431 digits by Gauss–Legendre quadrature of Pólya’s kernel, checked against $8\zeta(\frac{1}{2})$ and $\zeta''(\frac{1}{2})$, and ran the Euclidean algorithm in 420-digit arithmetic. These degrees are far below the range $d \leq 2 \cdot 10^{17}$ of [16] at $n = 0$; what the chains add is the size of the couplings and a control of the reach. Every coupling is positive, at full length, for all 7029 polynomials with $2 \leq d \leq 100$ and $0 \leq n \leq 700$ in steps of 10, and for $d = 80, 120, 160, 200, 260, 320, 400$ at $n = 0, 1, 10, 100, 300$. After scaling to unit mean square of the roots, the smallest coupling divided by the corresponding coupling $k/2$ of the Hermite chain, the limit of [24], is 0.765, at $d = 400, n = 0$ (Figure 6a).

The test has a reach, which a control measures. In the product $\prod_{k \leq 40} (1 + X/\gamma_k^2)$ over the first 40 zeros, all Jensen polynomials up to degree 10^7 are real-rooted. When one zero is moved off the line by δ , the first non-real degree is 237 for γ_1 with $\delta = 0.5$, 23714 for γ_1 with $\delta = 0.05$, 7499 for γ_{20} with

$\delta = 0.5$ and 749894 for γ_{20} with $\delta = 0.05$: about $0.3(\gamma/\delta)^2$ (Figure 6b). Degree 400 therefore certifies only $\gamma/\delta \lesssim 36$, far less than the verification of [46], and by the same scaling the degree $2 \cdot 10^{17}$ of [16] reaches γ/δ of order $8 \cdot 10^8$. The chains are an exact form of RH, not a test of it at large heights.

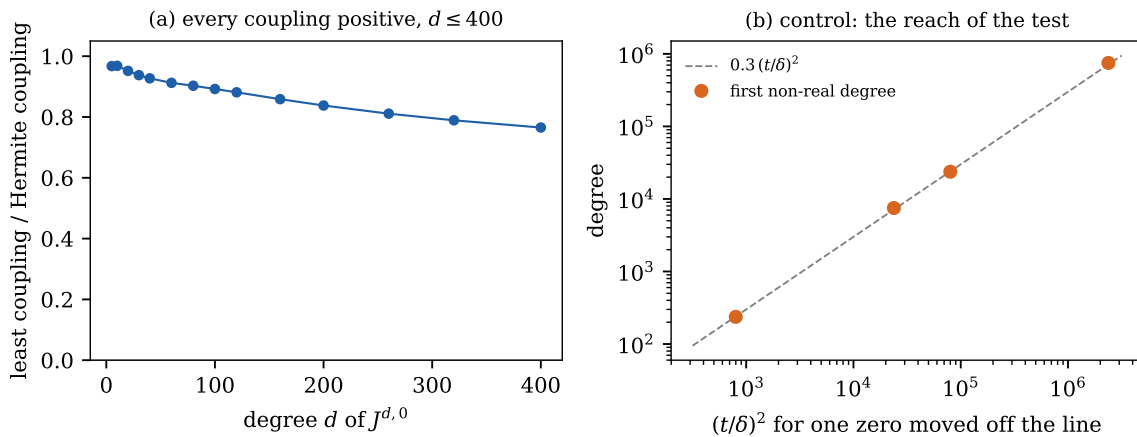


Figure 6. (a) The smallest coupling of the chain of $J^{d,0}$, relative to the Hermite chain, for $d \leq 400$: all positive. (b) Control: the degree at which a single zero moved off the line by δ is first detected, against $(\gamma/\delta)^2$.

9. The Reversed Pattern

Reversing the sign of every prime in the Euler product gives the reversed pattern $\prod_p (1 + p^{-s})^{-1}$. A zero of ζ , where the forward pattern goes to $-\infty$ in logarithm, is a pole of the reversed pattern, where it goes to $+\infty$: the same pattern read in reverse order.

Proof of Theorem 8. (i) $(1 + p^{-s})^{-1} = (1 - p^{-s})(1 - p^{-2s})^{-1} = \sum_{k \geq 0} (-1)^k p^{-ks}$, so the product is $\zeta(2s)/\zeta(s)$ and, λ being completely multiplicative with $\lambda(p) = -1$, equals $\sum \lambda(n)n^{-s}$ for $\sigma > 1$. If $\zeta(\rho) = 0$ with $\beta \geq \frac{1}{2}$ then $\text{Re}(2\rho) \geq 1$, so $\zeta(2\rho)$ is finite and nonzero [58, §3.3] and ρ is a pole of the same order. At $s = \frac{1}{2}$, $\zeta(2s)$ has a simple pole with residue $\frac{1}{2}$ and $\zeta(\frac{1}{2}) \neq 0$. (ii) The product $\zeta(s) \cdot \zeta(2s)/\zeta(s) = \zeta(2s)$ is the identity $\sum_{d|n} \lambda(d) = [n \text{ is a square}]$ [1, Theorem 2.19]. Hence $\sum_{n \leq x} L(x/n) = \sum_{dm \leq x} \lambda(d) = \sum_{k \leq x} \sum_{d|k} \lambda(d)$ counts the squares up to x . (iii) Let $m(y) = \sum_{n \leq y} \mu(n)/n$. For $y \geq 1$, $\sum_{n \leq y} \mu(n) \lfloor y/n \rfloor = 1$, so $ym(y) = 1 + \sum_{n \leq y} \mu(n) \{y/n\}$; the term $n = 1$ is $\{y\}$ and the others total at most $\lfloor y \rfloor - 1$ in absolute value, so $|ym(y)| \leq y$ and $|m(y)| \leq 1$. Also $m(y) \rightarrow 0$ [1, Ch. 4]. Since $\lambda(n) = \sum_{d^2|n} \mu(n/d^2)$,

$$\ell(N) = \sum_{d \leq \sqrt{N}} \frac{1}{d^2} m\left(\frac{N}{d^2}\right),$$

and dominated convergence, with the bound d^{-2} , gives $\ell(N) \rightarrow 0$. Next, $\sum_{k \leq x} \lambda(k) \lfloor x/k \rfloor = \sum_{n \leq x} L(x/n) = \lfloor \sqrt{x} \rfloor$ by (ii), so $\sum_{k \leq x} \lambda(k) \{x/k\} = x\ell(x) - \lfloor \sqrt{x} \rfloor$; for $x < k \leq N$, $\{x/k\} = x/k$, which adds $x(\ell(N) - \ell(x))$. This is (5) for $N \geq x$, and letting $N \rightarrow \infty$ gives the series. For $0 < x < 1$ both sides of (5) vanish in the limit. $\square$

In the variable $t = 1/x$, with $\varsigma_k(t) = \{1/(kt)\}$, (5) reads $-\sum_k \lambda(k) \varsigma_k(t) = \lfloor t^{-1/2} \rfloor$ for every $t > 0$. The forward counterpart is $-\sum_k \mu(k) \varsigma_k(t) = \chi_{(0,1]}(t)$, by (1) and $\sum_k \mu(k)/k = 0$ [1]: the Möbius-weighted sawtooths rebuild the unit step, with its single jump at $t = 1$, and the Liouville-weighted sawtooths rebuild the square staircase, with a jump at every $t = 1/n^2$.

Centre and Lean

The forward pattern $\zeta(s)$ has a single pole, at its centre $s = 1$. The reversed pattern has its centre pole at $s = \frac{1}{2}$, made by the squares through $\zeta(2s)$. In Perron's formula for $L(x)$ this pole contributes $\sqrt{x}/\zeta(\frac{1}{2}) = -0.6848\sqrt{x}$, and in that for $\ell(K)$, whose Dirichlet series is $\zeta(2+2w)/\zeta(1+w)$

at $w = s - 1$, it contributes $-1/(\zeta(\frac{1}{2})\sqrt{K}) = 0.6848/\sqrt{K}$. The measured mean of $L(x)/\sqrt{x}$ is -0.6834 (Table 7), and $\sqrt{K} \ell(K)$ leans to the positive side, while $\sqrt{N} m(N)$ is centred (Table 12). The other poles of the reversed pattern are the zeros of ζ . Every one computed lies on the vertical line through the centre $s = \frac{1}{2}$: the 2469 below height 3000 used here, and all below height $3 \cdot 10^{12}$ [46].

Table 12. Value imbalance of the first K terms of the reversed and forward patterns. The reversed one leans around $-1/\zeta(\frac{1}{2}) = 0.6848$, the forward one around 0.

K	10^2	10^3	10^4	10^5	10^6	10^7	10^8
$\sqrt{K} \ell(K)$ (reversed)	1.0787	0.9169	0.4280	0.4437	0.8144	1.0877	0.9960
$\sqrt{K} m(K)$ (forward)	+0.3113	+0.1395	-0.2083	-0.1541	+0.2006	+0.3210	+0.2047

Forward Times Reversed at a Zero

The identity of Theorem 8(ii) holds as an identity of meromorphic functions. At a zero of ζ its left side is a zero times a pole.

Proposition 1 (the identity at a zero). *Let $\rho = \beta + i\gamma$ be a zero of ζ of order m with $\beta \geq \frac{1}{2}$. Then:*

- $\zeta(2s)/\zeta(s)$ has a pole of order m at ρ ;
- the product $\zeta(s) \cdot \zeta(2s)/\zeta(s)$ is analytic at ρ , with the value $\zeta(2\rho)$, which is finite and nonzero;
- for $m = 1$,

$$\operatorname{Res}_{s=\rho} \frac{\zeta(2s)}{\zeta(s)} = \frac{\zeta(2\rho)}{\zeta'(\rho)}.$$

In $\sigma \geq \frac{1}{2}$ the squares pattern $\zeta(2s)$ has a single pole, at the centre $s = \frac{1}{2}$, where $\zeta(\frac{1}{2}) \neq 0$.

Proof. Write $\zeta(s) = (s - \rho)^m h(s)$ with h analytic near ρ and $h(\rho) \neq 0$. Then $\zeta(2s)/\zeta(s) = (s - \rho)^{-m} \zeta(2s)/h(s)$, and the product of the two patterns is $\zeta(2s)$. This is analytic at ρ because $\operatorname{Re}(2\rho) \geq 1$ and, γ being nonzero, $2\rho \neq 1$. Also $\zeta(2\rho) \neq 0$ by [58, §3.3]. For $m = 1$, $h(\rho) = \zeta'(\rho)$. The only pole of ζ is at 1, so the only pole of $\zeta(2s)$ is at $\frac{1}{2}$. $\square$

At every zero, the zero of the forward pattern and the pole of the reversed pattern cancel in the product. They cancel in the same way whether $\beta = \frac{1}{2}$ or $\beta > \frac{1}{2}$. For $\beta > \frac{1}{2}$ the point 2ρ lies in $\sigma > 1$. For $\beta = \frac{1}{2}$ it lies on the edge $\sigma = 1$ at height 2γ , away from the pole at 1. In both cases $\zeta(2\rho)$ is finite and nonzero. The identity therefore holds at every zero and does not by itself tell a zero on the line from a zero off it. The pole of the squares pattern sits at the centre alone, and the centre is not a zero of ζ . Table 13 gives the values at the first five zeros.

Table 13. The identity of Theorem 8(ii) at the first five zeros, computed with 30 significant digits. The last column gives the difference between $(s - \rho)\zeta(2s)/\zeta(s)$ at $s = \rho + 10^{-12}$ and $\zeta(2\rho)/\zeta'(\rho)$. At the same point the product $\zeta(s) \cdot \zeta(2s)/\zeta(s)$ differs from $\zeta(2\rho)$ by at most 2.3×10^{-12} ; both differences are of the size of the offset. With $|1 - 2^{-2i\gamma}|^2$, the values $|\zeta(1 + 2i\gamma)|^2$ give the edge weights $|\eta(1 + 2i\gamma)|^2$ of Section 10.

γ	$\zeta(2\rho) = \zeta(1 + 2i\gamma)$	$ \zeta(2\rho)/\zeta'(\rho) $	residue, direct
14.134725	1.836735 - 0.651198i	2.456952	2.1×10^{-12}
21.022040	0.812565 + 0.173883i	0.730941	2.7×10^{-13}
25.010858	0.444955 + 0.295572i	0.389423	1.1×10^{-12}
30.424876	0.374904 + 0.352748i	0.394777	1.4×10^{-12}
32.935062	0.789343 - 0.194577i	0.588206	3.1×10^{-13}

Checks

The identity (4) holds exactly at $x = 10, 99, 100, 1000, 12345$ and 10^5 , with values 3, 9, 10, 31, 111 and 316. Table 14 shows the partial sums of (5); by (5) they equal $\lfloor \sqrt{x} \rfloor - x \ell(K)$, and the direct sums over $k \leq 10^4$ agree with this to all digits shown. The exact sieve gives $L(10^8) = -3884$.

Table 14. The Liouville-weighted sawtooths rebuild the square staircase: $-\sum_{k \leq K} \lambda(k) \{x/k\}$ tends to $\lfloor \sqrt{x} \rfloor$.

x	$\lfloor \sqrt{x} \rfloor$	$K = 10^4$	$K = 10^6$	$K = 10^8$
0.7	0	-0.00300	-0.00057	-0.00007
3.9	1	0.98331	0.99682	0.99961
50.5	7	6.78385	6.95887	6.99497
99.99	9	8.57202	8.91856	8.99004
1000.3	31	26.71844	30.18532	30.90037

10. The Reversed Staircase Criterion

The criterion of Nyman and Beurling [9,41], in the form of Báez-Duarte [2], reads RH from the unit step $\chi_{(0,1]}$, which the Möbius-weighted sawtooths rebuild exactly: RH holds if and only if $d_N \rightarrow 0$. The reversed pattern rebuilds the square staircase (Theorem 8(iii)), and its alternating form f gives a second sawtooth criterion, built from the squares, in the direction that matters for RH: if $D_N \rightarrow 0$ then RH holds.

Lemma 1. (a) $\int_0^\infty \zeta_k(t) t^{s-1} dt = -\zeta(s) k^{-s} / s$ for $0 < \sigma < 1$.
 (b) (Vasyunin [59]) For $h, k \geq 1$ with $d = \gcd(h, k)$, $h = dh_1$, $k = dk_1$,

$$\langle \zeta_h, \zeta_k \rangle = \frac{1}{d} \left[\frac{\log 2\pi - \gamma_E}{2} \left(\frac{1}{h_1} + \frac{1}{k_1} \right) + \frac{k_1 - h_1}{2h_1k_1} \log \frac{h_1}{k_1} - \frac{\pi}{2h_1k_1} (V(h_1, k_1) + V(k_1, h_1)) \right],$$

with $V(h, k) = \sum_{m=1}^{k-1} \{mh/k\} \cot(\pi m/k)$.

(c) $\langle f, \zeta_k \rangle = \int_1^\infty [\lfloor \sqrt{u} \rfloor \text{ odd}] \{u/k\} u^{-2} du$, and on an interval $[y, y']$ containing no multiple of k and no square in its interior, where $\{u/k\} = (\alpha + u - y)/k$ with $\alpha = k\{y/k\}$, the integrand integrates to $k^{-1}(\log(y'/y) - (y - \alpha)(y' - y)/(yy'))$. For the unit step, $\langle \chi_{(0,1]}, \zeta_k \rangle = (\log k + 1 - \gamma_E)/k$.

Proof. (a) Substituting $y = 1/(kt)$ gives $k^{-s} \int_0^\infty \{y\} y^{-s-1} dy = -k^{-s} \zeta(s)/s$ [58, §2.1]. (c) Substitute $u = 1/t$; then $\int_y^{y'} (\alpha - y + u) u^{-2} du = \log(y'/y) - (y - \alpha)(\frac{1}{y} - \frac{1}{y'})$. $\square$

Proof of Theorem 9. (i) $f = 1$ exactly on the intervals $((n+1)^{-2}, n^{-2}]$ with n odd, so $\|f\|^2 = \sum_{n \text{ odd}} (n^{-2} - (n+1)^{-2}) = \sum_{m \geq 1} (-1)^{m-1} m^{-2} = \eta(2)$, and for $\sigma > 0$, $\int_0^\infty f(t) t^{s-1} dt = s^{-1} \sum_{n \text{ odd}} (n^{-2s} - (n+1)^{-2s}) = \eta(2s)/s$. By (5) at $x = 1/t$ and at $x/4$,

$$-\sum_k \lambda(k) \{x/k\} + 2 \sum_m \lambda(m) \{x/(4m)\} = \lfloor \sqrt{x} \rfloor - 2 \lfloor \sqrt{x}/2 \rfloor = [\lfloor \sqrt{x} \rfloor \text{ odd}],$$

since $\lfloor \sqrt{x}/2 \rfloor = \lfloor \lfloor \sqrt{x} \rfloor / 2 \rfloor$; this is the stated identity, both series converging. The weights have the Dirichlet series $\zeta(2s)/\zeta(s) - 2 \cdot 4^{-s} \zeta(2s)/\zeta(s) = (1 - 2^{1-2s}) \zeta(2s)/\zeta(s) = \eta(2s)/\zeta(s)$.

(ii) Let $c^{(N)}$ attain D_N , $g_N = f - \sum_{k \leq N} c_k^{(N)} \zeta_k$ and $A_N(s) = \sum_{k \leq N} c_k^{(N)} k^{-s}$. For $t > 1$, $\zeta_k(t) = 1/(kt)$ and $f(t) = 0$, so $g_N(t) = -A_N(1)/t$ and $D_N^2 \geq \int_1^\infty A_N(1)^2 t^{-2} dt = A_N(1)^2$; hence $A_N(1) \rightarrow 0$. Suppose $\zeta(s_0) = 0$ with $\frac{1}{2} < \sigma_0 < 1$. By (i) and Lemma 1(a) the absolutely convergent Mellin integral of g_N at s_0 is

$$\int_0^\infty g_N(t) t^{s_0-1} dt = \frac{\eta(2s_0)}{s_0} + \frac{\zeta(s_0) A_N(s_0)}{s_0} = \frac{\eta(2s_0)}{s_0}.$$

The part over $(1, \infty)$ is $-A_N(1) \int_1^\infty t^{s_0-2} dt = -A_N(1)/(1 - s_0) \rightarrow 0$, and by Cauchy-Schwarz the part over $(0, 1]$ is at most $D_N(2\sigma_0 - 1)^{-1/2} \rightarrow 0$. Hence $\eta(2s_0) = 0$. But $\eta(2s) = (1 - 2^{1-2s}) \zeta(2s)$, where $|2^{1-2s}| = 2^{1-2\sigma} < 1$ and $\zeta(2s) \neq 0$ by the Euler product for $\sigma > \frac{1}{2}$. So $\zeta(s) \neq 0$ for $\frac{1}{2} < \sigma < 1$; with $\zeta(s) \neq 0$ for $\sigma \geq 1$ [58, §3.3] and the functional equation, every zero in the strip lies on $\sigma = \frac{1}{2}$.

(iii) Let $g = f - \sum_{k \leq N} c_k \zeta_k$. On $(0, 1]$, $|g| \leq 1 + \sum |c_k|$, and on $(1, \infty)$, $|g(t)| = |A(1)|/t$, so the Mellin integral of g converges absolutely on $\sigma = \frac{1}{2}$, and by (i) and Lemma 1(a) it equals $\eta(2s)/s + \zeta(s)A(s)/s$, which is $\eta(2\rho)/\rho$ at a zero ρ . For the unit step the transform of $\chi_{(0,1]}$ is $1/s$. $\square$

The Norm and the Value at -1

The functional equation $\zeta(1-s) = 2(2\pi)^{-s} \cos \frac{\pi s}{2} \Gamma(s) \zeta(s)$ [58, §2.1] at $s = 2$ reflects $s = 2$ through the midpoint $s = \frac{1}{2}$ to $s = -1$ and gives $\zeta(-1) = -\zeta(2)/2\pi^2$. Since $\zeta(2) = 2\eta(2)$, Theorem 9(i) gives

$$\zeta(-1) = -\frac{\|f\|^2}{\pi^2} = -\frac{1}{12} :$$

the value that Ramanujan assigned to $1 + 2 + 3 + \dots$ [5, Ch. 6], [30] is the norm of the odd-square staircase, reflected through the midpoint and divided by $-\pi^2$.

The Weight of a Zero

By the Mellin–Plancherel formula $\|g\|^2 = \frac{1}{2\pi} \int_{-\infty}^{\infty} |\hat{g}(\frac{1}{2} + it)|^2 dt$, and by Theorem 9(iii) no choice of coefficients moves the transform of the error at a zero on the line: it stays $1/\rho$ for the unit step and $\eta(1 + 2i\gamma)/\rho$ for the odd-square staircase. The reversed reading therefore weights each zero on the line by the value of η on the edge $\sigma = 1$ at twice its height. For the unit step, Burnol [15] proved that the pinned values give $\liminf_N d_N^2 \log N \geq \sum_{\text{Re } \rho=1/2} \text{mult}(\rho)^2 / |\rho|^2$, and Bettin, Conrey and Farmer [6] proved $d_N^2 \sim C_{\text{fwd}} / \log N$ under RH and a hypothesis on $\sum_{\rho} 1/|\zeta'(\rho)|^2$, with

$$C_{\text{fwd}} = \sum_{\rho} \frac{1}{\rho(1-\rho)} = 2 + \gamma_E - \log 4\pi = 0.0461914,$$

which is $\sum_{\rho} 1/|\rho|^2$ when every $\beta = \frac{1}{2}$, the setting of [6]. The reversed pinned values attach to the zeros on the line the constant

$$C_{\text{rev}} = \sum_{\text{Re } \rho=1/2} \frac{|\eta(2\rho)|^2}{|\rho|^2} = 0.075016,$$

computed over the 2469 zeros with $0 < \gamma < 3000$ and both signs of γ , completed by the tail beyond 3000 with the density of the zeros and the mean value $\zeta(2) = \pi^2/6$ of $|\eta(1 + iu)|^2$; the ratio is $C_{\text{rev}}/C_{\text{fwd}} = 1.624$. The edge weights $|\eta(1 + 2i\gamma)|^2$ of the first ten zeros are 2.014, 2.273, 1.138, 0.653, 1.460, 2.229, 0.115, 3.692, 1.083 and 0.026.

10.1. Computation

For $N \leq 60$ the Gram matrix $\langle \zeta_h, \zeta_k \rangle$ is computed by exact integration on the intervals where both sawtooths are linear; for $N \leq 2000$ by Lemma 1(b), which agrees with the exact integrals to 3.1×10^{-13} on the 60×60 block. The inner products $\langle f, \zeta_k \rangle$ are computed by Lemma 1(c), splitting $[1, 2000^2]$ at the multiples of k and at the squares, with the tail beyond 2000^2 evaluated with the mean $\frac{1}{2}$ of $\{u/k\}$; the same procedure gives $\langle \chi_{(0,1]}, \zeta_k \rangle$ to within 2.6×10^{-13} of $(\log k + 1 - \gamma_E)/k$, and direct quadrature on a truncated range agrees to 10^{-15} . Then $D_N^2 = \eta(2) - \mathbf{b}^T G^{-1} \mathbf{b}$, with $\mathbf{b} = (\langle f, \zeta_k \rangle)_{k \leq N}$, by a Cholesky factorisation of G , whose condition number is 1.27×10^4 at $N = 60$, 3.84×10^6 at $N = 1000$ and 1.50×10^7 at $N = 2000$.

In double precision, with unit roundoff $u = 1.1 \times 10^{-16}$, the condition number bounds the relative error of the Cholesky solution by a quantity of order $1.50 \times 10^7 u \approx 2 \times 10^{-9}$, far below the digits shown. A second implementation of the same formulas reproduces the condition number and the values of D_N^2 in Tables 16 and 17 at the 15 values of N recomputed. The matrix serves only to compute D_N at finite N . Theorem 9(ii) does not use it: the theorem holds for any coefficients that attain D_N .

The Exact Weights Cut at N

The weights $w_k = \lambda(k) - 2\lambda(k/4)[4 \mid k]$ of Theorem 9(i) rebuild f at every point. They also satisfy $\sum_k w_k/k = 0$, the value at $s = 1$ of $\eta(2s)/\zeta(s)$, where ζ has its pole. This follows from Theorem 8(iii): $\sum_k \lambda(k)/k = 0$, and $\sum_{4 \mid k} \lambda(k/4)/k = \frac{1}{4} \sum_m \lambda(m)/m = 0$. Cut at N , the weights give the error

$$g_N = f + \sum_{k \leq N} w_k \zeta_k = - \sum_{k > N} w_k \zeta_k.$$

For $t > 1$, $g_N(t) = W_N/t$ with $W_N = \sum_{k \leq N} w_k/k = -\sum_{k > N} w_k/k$. So the part of $\|g_N\|^2$ over $(1, \infty)$ is W_N^2 , which tends to 0; this is the first step of the proof of Theorem 9(ii). The part over $(0, 1]$, where the sawtooths are not linear, is not controlled by W_N . Table 15 computes both parts with the same Gram matrix, $\|g_N\|^2 = \eta(2) + 2 \sum_k w_k \langle f, \varsigma_k \rangle + \mathbf{w}^T \mathbf{G} \mathbf{w}$.

Table 15. The odd-square staircase approximated by its exact weights cut at N , against the distance D_N^2 attained by the best coefficients. The error of the cut weights over $(1, \infty)$ is W_N^2 . Over $(0, 1]$ it does not decrease.

N	D_N^2	W_N^2 , part over $(1, \infty)$	part over $(0, 1]$	$\sum_{k \leq N} w_k$
10	0.052277	5.87×10^{-3}	0.14267	0
30	0.023911	1.03×10^{-3}	0.09117	-2
60	0.021086	3.27×10^{-4}	0.08574	0
100	0.017627	1.62×10^{-4}	0.07427	0
200	0.015199	2.64×10^{-4}	0.12457	-4
300	0.013664	2.32×10^{-5}	0.06351	-2
500	0.012777	7.39×10^{-6}	0.06715	-2
700	0.011496	4.51×10^{-7}	0.05846	0
1000	0.011124	4.64×10^{-7}	0.06261	-2
1400	0.010606	4.51×10^{-5}	0.13060	8
2000	0.010119	9.05×10^{-5}	0.28049	18

Over $300 \leq N \leq 2000$ the part over $(0, 1]$ lies between 0.058 and 0.281 and does not decrease, while D_N^2 falls to 0.0101. It is largest where the partial sum $\sum_{k \leq N} w_k$ is largest: +8 at $N = 1400$ and +18 at $N = 2000$. The exact weights cut at N rebuild f at every point but not in the norm. The vanishing of $\eta(2s)/\zeta(s)$ at the pole of ζ settles the distance over $t > 1$. The distance over $0 < t \leq 1$ is carried by the coefficients that attain D_N , which are chosen afresh at each N . By Theorem 9(ii) the statement $D_N \rightarrow 0$ gives RH, and it is a statement about those coefficients.

Table 16. The distances of the odd-square staircase (D_N) and of the unit step (d_N) from the span of N sawtooths, for $N \leq 60$.

N	1	2	3	4	5	6	8	10	12
D_N^2	.731629	.253830	.155946	.130060	.129492	.104097	.072527	.052277	.050267
d_N^2	.858212	.289965	.093837	.065129	.036102	.035986	.024161	.023772	.020443
N	15	20	25	30	40	45	50	55	60
D_N^2	.046176	.030875	.026741	.023911	.022696	.022493	.022021	.021487	.021086
d_N^2	.018335	.016504	.015520	.014530	.012715	.012023	.011869	.011623	.011471

Table 17. Distances times $\log N$ up to $N = 2000$, against the constants $C_{\text{fwd}} = 0.04619$ and $C_{\text{rev}} = 0.07502$ of the zeros. Over $1200 \leq N \leq 2000$ the means lie 0.6% below C_{fwd} and 2.5% above C_{rev} ; the ratio at $N = 2000$ is 1.655, against $C_{\text{rev}}/C_{\text{fwd}} = 1.624$.

N	$d_N^2 \log N$	$D_N^2 \log N$	ratio	window of N	mean $d_N^2 \log N$	mean $D_N^2 \log N$
60	0.04697	0.08634	1.838	40–100	0.04720	0.08600
100	0.04692	0.08118	1.730	100–300	0.04650	0.07956
200	0.04726	0.08053	1.704	300–700	0.04582	0.07850
300	0.04560	0.07794	1.709	700–1200	0.04514	0.07640
400	0.04618	0.07822	1.694	1200–2000	0.04591	0.07686
500	0.04611	0.07940	1.722	constant	0.04619	0.07502
700	0.04421	0.07531	1.704			
1000	0.04529	0.07684	1.697			
1400	0.04571	0.07683	1.681			
2000	0.04647	0.07691	1.655			

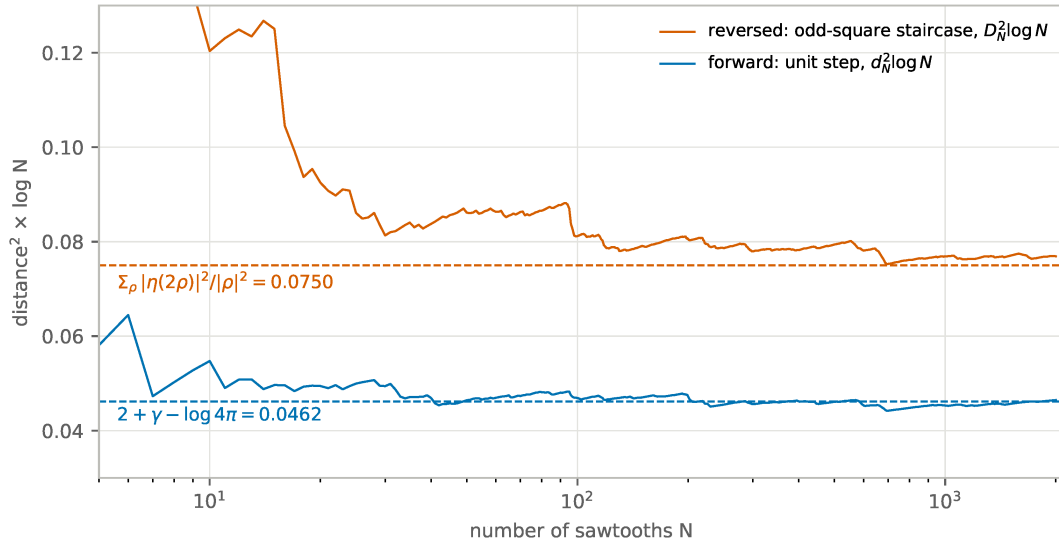


Figure 7. $D_N^2 \log N$ for the odd-square staircase and $d_N^2 \log N$ for the unit step, $5 \leq N \leq 2000$, computed from exact integrals, with the constants C_{rev} and C_{fwd} of the zeros (dashed).

11. Prime Clocks

The clock of a prime p is $t \mapsto p^{it}$; at height t it reads $t \log p / 2\pi$ turns. At a whole turn the clock is aligned, $p^{it} = 1$; at a half turn it is reversed, $p^{it} = -1$. At $t = 0$ every clock is aligned and the Euler product $\prod_p (1 - p^{-s})^{-1}$ has its pole at $s = 1$; turning every clock by half a turn, $p^{-s} \mapsto -p^{-s}$, gives the reversed pattern of Section 9. The pitch in doublings of the wave of a zero (Section 2) is the reading of the clock of 2 at its height.

Proof of Theorem 10. The first statement is Landau's formula [33]; Gonek [26] gives a version uniform in x . When $\rho = \frac{1}{2} + i\gamma$, $x^\rho = \sqrt{x} e^{i\gamma \log x}$, and taking real parts gives the mean of $\cos(\gamma \log x)$. For the clock of p , $\frac{1}{N(T)} \sum e^{2\pi i k \gamma \log p / 2\pi} = \frac{1}{N(T)} \sum p^{ik\gamma} = \frac{p^{-k/2}}{N(T)} \sum (p^k)^\rho$, and $\Lambda(p^k) = \log p$. $\square$

With these Fourier coefficients, the density of the readings of the clock of p over the zeros up to height T is

$$\delta_p(\phi) = 1 - \frac{T \log p}{\pi N(T)} \sum_{k \geq 1} p^{-k/2} \cos 2\pi k \phi, \quad (12)$$

with $\delta_p(0) = 1 - \frac{T \log p}{\pi N(T)(\sqrt{p}-1)}$ at the alignment and $\delta_p(\frac{1}{2}) = 1 + \frac{T \log p}{\pi N(T)(\sqrt{p}+1)}$ at the reversal; this is the main term, each coefficient carrying an error $O(\log T / N(T))$. Every coefficient is negative: each prime pulls the zeros toward the reversal of its clock, with strength $\Lambda(x) / \sqrt{x}$ at $x = p^k$. The zeros sit on the reversed side of every clock. For $p = 2$, $T = 3000$ and $N(T) = 2469$, $\delta_2(0) = 0.353$ and $\delta_2(\frac{1}{2}) = 1.111$.

11.1. Measurements

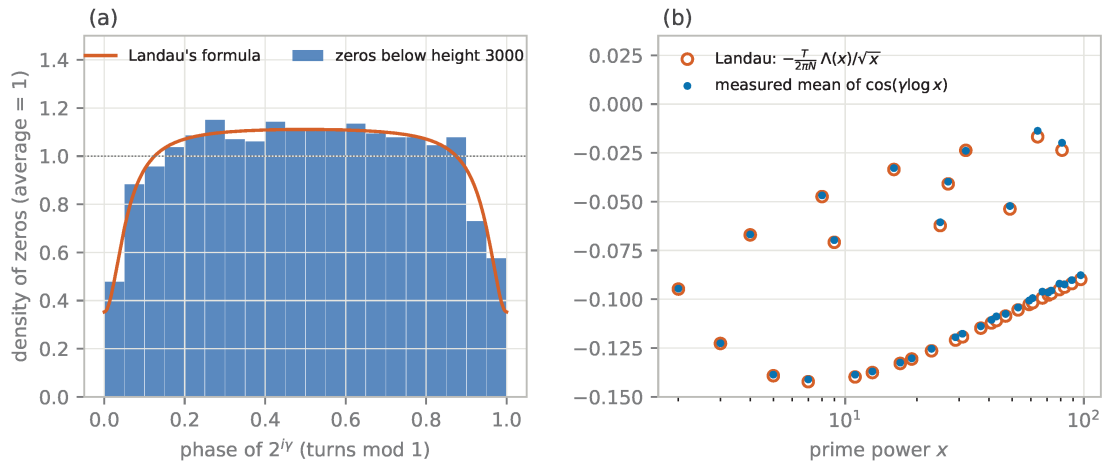
Over the 2469 zeros with $0 < \gamma < 3000$, the mean of $\cos(\gamma \log x)$ agrees with Landau's formula to within 0.0012 at every prime power up to 13 (Table 18) and to within 0.0038 at all 35 prime powers up to 100, the largest difference being at $x = 81$. The density of the zeros on the clock of 2 follows (12) (Table 19, Figure 8): near the alignment it is 0.68 and 0.65 of the average, and elsewhere about 1.1.

Table 18. Mean of $\cos(\gamma \log x)$ over the 2469 zeros below height 3000, and $-\frac{T}{2\pi N(T)} \Lambda(x) / \sqrt{x}$.

x	2	3	4	5	7	8	9	11	13
measured	−.0946	−.1225	−.0668	−.1386	−.1411	−.0468	−.0697	−.1387	−.1370
Landau	−.0948	−.1227	−.0670	−.1392	−.1422	−.0474	−.0708	−.1398	−.1376

Table 19. Density of the zeros below height 3000 on the clock of 2, per tenth of a turn (average 1).

reading mod 1	0–.1	.1–.2	.2–.3	.3–.4	.4–.5	.5–.6	.6–.7	.7–.8	.8–.9	.9–1
measured	0.680	0.996	1.118	1.065	1.126	1.110	1.114	1.077	1.061	0.652
by (12)	0.671	1.026	1.088	1.105	1.110	1.110	1.105	1.088	1.026	0.671

**Figure 8.** (a) Density of the 2469 zeros below height 3000 on the clock of 2, in twentieths of a turn, and the density (12). (b) Mean of $\cos(\gamma \log x)$ over the same zeros at the 35 prime powers $x \leq 100$, and Landau's formula.

The Clock of 2 in the Triad Charge

Since $\chi_{-3}(2) = -1$, the Euler factor of $L(s, \chi_{-3})$ at 2 is $(1 + 2^{-s})^{-1}$, whose modulus at $s = \frac{1}{2} + i\gamma$ is $|1 + 2^{-1/2} 2^{-i\gamma}|^{-1}$: 3.414 at a half turn of the clock of 2 and 0.586 at a whole turn. The wave amplitudes of the triad charge $a(x)$ in (6) carry $|L(\rho, \chi_{-3})|$, and over the zeros below height 3000 the mean of $|L(\rho, \chi_{-3})|$ is 4.24 for the 274 zeros within 0.05 turn of a half turn and 0.98 for the 130 zeros within 0.05 turn of a whole turn; divided by the factor at 2, the two means are 1.38 and 1.66. The sixth zero, whose wave is nearly silent in the triad charge (Table 3), reads 4.1464 turns, 0.15 turn past a whole turn.

11.2. The Tenth Zero

The tenth zero $\gamma_{10} = 49.7738$ reads 5.4909 turns on the clock of 2, within 0.009 of $5\frac{1}{2}$, which in base two is 101.1; there $2^{i\gamma_{10}} = -0.9984 + 0.0569i$, and the clock is reversed. The reversed reading of Theorem 9(iii) weights the zero by

$$|\eta(1 + 2i\gamma)|^2 = |1 - 2^{-2i\gamma}|^2 |\zeta(1 + 2i\gamma)|^2, \quad |1 - 2^{-2i\gamma}| = 2|\sin(\gamma \log 2)|,$$

and at twice the height the clock of 2 reads 10.9819 turns, within 0.018 of 11, which in base two is 1011: doubling the height moves the binary point one place, the half turn becomes a whole number of turns, and the factor $1 - 2^{1-2s}$ of $\eta(2s)$ closes. For γ_{10} , $|1 - 2^{-2i\gamma}|^2 = 0.0129$, $|\zeta(1 + 2i\gamma)|^2 = 2.04$, and the weight is 0.0264, against a mean of 1.52 over the 2469 zeros.

The factor $|1 - 2^{-2i\gamma}|^2$ vanishes whenever the clock of 2 reads a half or a whole turn. of the 2469 zeros below height 3000, 71 have weight below 0.05: 49 lie within 0.1 turn of a half turn and 22 within 0.1 turn of a whole turn, more of them near the reversal, where the density (12) is higher. The first seven are, as (rank, γ , reading mod 1, weight), (10, 49.774, 0.491, 0.0264), (124, 276.452, 0.498, 0.0029), (163, 339.858, 0.492, 0.0263), (172, 353.489, 0.996, 0.0111), (269, 498.581, 0.002, 0.0016), (311, 557.565, 0.509, 0.0451) and (327, 580.137, 0.999, 0.0004).

Remark 3 (rank and value). *The silence of the tenth zero in the reversed reading is a property of its value, the height γ_{10} read on the clock of the prime 2, and not of its rank: the same clock silences 70 further zeros below height 3000. That prime is 2, written 10 in base two.*

12. Methods of Computation

Counts to 10^{10}

An exact segmented sieve gives $\mu(n)$, $\lambda(n)$ and $\Lambda(n)$ to 10^{10} and records π , M , L , θ , ψ , their restrictions to the classes 1 and 5 mod 6, and $\pi - \text{li}$ in extended precision at 675,247 checkpoints; it gives $\pi(10^{10}) = 455,052,511$, $M(10^{10}) = -33,722$ and $L(10^{10}) = -116,026$. The grid $u = 12 + 0.001j$ is interpolated linearly between checkpoints; J is formed from π and a table of $\pi(y)$ for $y \leq 10^5$.

The Tower

A second segmented sieve to 10^{10} computes for every n the values $\mu(n)$, $(\mu * \chi_{-3})(n)$ and $(\mu * \chi_{-4})(n)$ in integers and $(\mu * \chi_9)(n)$ and $(\mu * \chi_9 \chi_{-3})(n)$ in $\mathbb{Z}[\omega]$, stored as integer pairs (a_1, a_2) for $a_1 + a_2\omega$ with $\omega^2 = -1 - \omega$, from the values at prime powers in Theorem 2. Each n is divided exactly by the primes up to 10^5 , and the cofactor left after these divisions is 1 or a prime. All partial sums are exact and are recorded at 123,450 checkpoints, which contain the grid, the points $x/3$ and the values $\lfloor 10^{10}/r \rfloor$, $r \leq 10^5$. The leaves $M(10^{10}/r)$ come from these checkpoints for $r \leq 10^5$ and from a table of $M(v)$, $v \leq 10^5$, for $r > 10^5$, and give every class of the tower at every level exactly.

Spectra and Energies

Mean over the grid removed, Hann window, direct evaluation of the Fourier sum at each pitch.

Zeros and Special Values

The 2469 zeros with $0 < \gamma < 3000$ are computed with mpmath [38], whose count $N(3000) = 2469$ by Turing's method agrees; all lie on the critical line [46]. The functions $\zeta(s)$, $\zeta'(s)$ and $\zeta(s, a)$ are evaluated by Euler–Maclaurin summation, $L(s, \chi_{-3}) = 3^{-s}(\zeta(s, \frac{1}{3}) - \zeta(s, \frac{2}{3}))$ and $L(s, \chi_{-4}) = 4^{-s}(\zeta(s, \frac{1}{4}) - \zeta(s, \frac{3}{4}))$. The squares part is evaluated by (9) as $\log |E(s)| = -\sum_{2 \leq k \leq K} \mu(k)k^{-1} \log |\zeta(ks)|$, with $K = 24$ at the random points and $K = 79$ for Table 8.

Squared Pair

Q_ζ on the grid $\sigma = 0.52, 0.53, \dots, 1.10$, $t = 2 + 0.05j$ for Figure 3(a); the reach by 14 bisection steps in σ , with the minimum over 13 equally spaced heights in $[\gamma - 0.3, \gamma + 0.3]$.

Value Distribution

$\zeta(\sigma + it)$ at 6000 uniform random heights in $[5000, 6000]$; the model as described in Section 6; the extremes on the grid $t = 10 + 0.02j$.

Reversal

An exact sieve of λ to 10^8 gives $\ell(K)$ in double precision; (4) is checked by direct summation of L .

Sawtooth Integrals

As in Section 10; linear algebra in double precision by Cholesky factorisation. For the recomputation and Table 15, $\langle f, \varsigma_k \rangle = \sum_{n \text{ odd}} (\Phi_k((n+1)^2) - \Phi_k(n^2))$ with $\Phi_k(y) = \int_1^y \{u/k\} u^{-2} du = (\log y - H_{\lfloor y/k \rfloor})/k + \lfloor y/k \rfloor / y$, where H_m is the harmonic number. The sum runs over $n < 2 \cdot 10^5$, and the tail is evaluated with the mean $\frac{1}{2}$ of $\{u/k\}$.

The Identity at a Zero

The values of Table 13 are computed with 30 significant digits by the mpmath library [38].

Gaps and Classes

A sieve of the primes to 10^8 ; the window law is checked directly for $p \leq 19$.

Truncated Pair

Z_N evaluated in double precision with $\log |\chi|$ from the complex log-gamma function; for each N , the local minima of $|Z_N|$ on the grid $\sigma = 0.5005 + 0.005j$, $t = 2\pi N^2 + 0.01k$, below a fifth of the median, refined by Newton's method; ζ at the points and on the boxes with mpmath at 15 digits [38]. The list of located zeros is not claimed to be complete.

Jensen Chains

$\gamma(n)$ from $\gamma(n) = \frac{64 \cdot 4^n n!}{(2n)!} \int_0^\infty u^{2n} \Phi(u) du$ with Pólya's kernel $\Phi(u) = \sum_{k \geq 1} (2\pi^2 k^4 e^{9u} - 3\pi k^2 e^{5u}) e^{-\pi k^2 e^{4u}}$, by composite Gauss–Legendre quadrature on $[0, 2]$; the Euclidean algorithm on the polynomial centred at the mean of its roots and scaled to unit mean square, in mpmath at 420 digits.

13. Concluding Remarks

The Triad Tower

The rebuild law splits by the residue of the divisor into the self-similar tower of Theorem 1: its zero classes repeat the whole law one level down, its unit classes are the character sums G_χ , carried only by the primes that the characters do not fix (Theorem 2), and its leaves are the values $M(x/r)$, whose energy is subpolynomially above x exactly when RH holds. At every zero of ζ the triad of Dirichlet series is $(0, a, -a)$ with $a = \frac{1}{2}L(\rho, \chi_{-3})$ (Theorem 3), and this fixes every wave of the tower: at 10^{10} the class values agree exactly with the leaves, and the twelve leading wave amplitudes of M , a and b and their energies agree with the zeros to within 0.0003 and 1%.

The Squares

Each prime square opens the next wheel (Theorem 4): with 25 and 49 inserted, the primes to 61 are the wheels of 6 and 30, and the classes $6k \pm 1$ stand 9 : 9. All squares fall in the class $6k + 1$, and the logarithm of the Euler product counts each of them with weight $\frac{1}{2}$ (Theorem 5). At that weight the lean of π against li vanishes, with fitted weight 0.4994, and the lean of the classes $6k \pm 1$ drops from 1.157 to 0.031, with fitted weight 0.514; the lean of L sits in its square multiples, and without them L becomes M , which is centred. The waves are unchanged.

The Horizon

The squares part E of ζ is analytic, zero-free and satisfies the squared-pair inequality all the way down to the square-root horizon $\sigma = \frac{1}{2}$, where it has a half-order pole; the prime factor $F = \zeta/E$ carries every zero of ζ in $\sigma > \frac{1}{2}$ and has at $s = \frac{1}{2}$ the matching half-order zero, which is the square lean $-\frac{1}{2} \text{li}(\sqrt{x})$ of the prime count (Theorem 6, Remark 1). The phases $t \log p$ are jointly equidistributed on every finite torus, and in the model with independent phases the prime term has the squares series as variance and converges almost surely exactly beyond the horizon (Theorem 7); for each fixed $\sigma > \frac{1}{2}$ the zeros with $\beta > \sigma$ form a proportion $O(1/\log T)$ of those up to height T ; and the values of ζ at heights 5000 to 6000 follow the independent-phase law.

The Axiom

Harmonic independence is the absence of an integer relation (Axiom 1). By unique factorisation the frequencies $\log p$ of the prime waves admit none, which is a theorem (Section 7); the waves are related only through the Euler product that constructs ζ , and a zero off the critical line would be a relation without a mechanism.

The Reversed Pattern

Reversing every prime turns $\zeta(s)$ into $\zeta(2s)/\zeta(s)$ and every zero into a pole; forward times reversed is $\zeta(2s)$; and the Liouville-weighted sawtooths rebuild the square staircase $\lfloor \sqrt{x} \rfloor$ exactly (Theorem 8). The forward pattern has its single pole at its centre $s = 1$. The reversed pattern has

its centre pole at $s = \frac{1}{2}$, which gives $L(x)$ its lean $\sqrt{x}/\zeta(\frac{1}{2})$, and its other poles at the zeros of ζ . In this form the Riemann Hypothesis is the statement that every pole of the reversed pattern lies on the vertical line through its centre. The results bear on this statement from four sides. The squares part is positive and zero-free down to that line (Theorem 6). In the independent-phase model of the prime factor, which carries the poles, the prime term converges almost surely beyond the line with the squares series as variance, and for each fixed $\sigma > \frac{1}{2}$ the poles with $\beta > \sigma$ form a proportion $O(1/\log T)$ of those up to height T (Theorem 7). The odd-square staircase, rebuilt exactly by the reversed sawtooths, gives the criterion that $D_N \rightarrow 0$ puts every pole on the line, and computed to $N = 2000$, $D_N^2 \log N$ stays within 3.3% of the constant C_{rev} of the poles on the line over $1200 \leq N \leq 2000$ (Theorem 9, Table 17). Every prime clock pulls the poles toward its reversal as Landau's formula prescribes, as verified over the 2469 poles below height 3000 at the 35 prime powers up to 100 and in the density on the clock of 2 (Theorem 10, Tables 18 and 19). At each zero the forward zero and the reversed pole cancel in the product, leaving $\zeta(2\rho)$ with residue $\zeta(2\rho)/\zeta'(\rho)$. They cancel in the same way on the line and off it, so the identity holds at every zero and does not place it (Proposition 1). The exact weights of the staircase settle the distance over $t > 1$ through the vanishing of $\eta(2s)/\zeta(s)$ at $s = 1$. Over $0 < t \leq 1$ they leave an error that does not decrease up to $N = 2000$ (Table 15), so the criterion rests on the coefficients that attain D_N .

The Midpoint

The forward terms and their reversed partners are balanced, all at once, exactly on the critical line (Theorem 11). Without the remainder, the pair meets itself off the line: we locate 116 such zeros below height 5952, all at heights where the prime clocks are anti-aligned, and the complete ζ is nonzero at every one of them. With the remainder, every zero computed lies on the line. As positive chains, the Jensen polynomials of Ξ are real-rooted wherever tested, and a control shows how far such a test reaches. In the reading of Axiom 1, a zero is a meeting of the completed pattern with its reversal, and the construction supplies balance at the midpoint only.

Outlook

The computations extend directly: the tower beyond 10^{10} by the rebuild law, the reversed distance D_N beyond $N = 2000$ by Vasyunin's closed form, and the densities of the zeros on the clocks of the odd primes by (12).

14. Data and Sources

External Inputs

All inputs are mathematical; no measured data are used.

- (i) All nontrivial zeros of ζ with $0 < \gamma \leq 3 \cdot 10^{12}$ lie on the critical line [46].
- (ii) Results reported in 2026 and cited for context only: [4,18,44,45].
- (iii) Published values used for comparison: $\pi(10^n)$ [42]; $M(10^{10}) = -33,722$ [43]; the least $x > 1$ with $L(x) > 0$, 906,150,257 [57].
- (iv) Classical and published theorems: the rebuild law [1, Ch. 3]; Littlewood's criterion and the explicit formula for M [58, Theorems 14.25(C), 14.27]; Riemann's formula for J [51,62]; the prime number theorem and the explicit formula for $\psi(x, \chi)$ [21, Ch. 18, 19]; the Hurwitz zeta function, $\sum_{d|n} \lambda(d)$ and $\sum \mu(n)/n = 0$ [1]; Kronecker's theorem and Weyl's criterion [31,61]; the variance criterion and the three-series theorem [23,32]; the mean square of ζ [58, Theorem 7.2] and its second term [36]; the density theorems of Bohr and Landau and of Selberg [12, 54], [58, Theorems 9.15(A), 9.19(C)]; the value distribution [10,11]; Mertens' inequality [37], [58, §3.3, (3.3.1)]; Chebyshev's bias [53]; the sign changes of $\pi - \text{li}$ [35]; Pólya's question [50]; the criterion of Nyman, Beurling and Báez-Duarte [2,9,41]; Vasyunin's closed form [59]; Burnol's bound [15]; the asymptotic of Bettin, Conrey and Farmer [6]; Landau's formula [26,33]; the approximate functional equation [56], [58, §4.12]; the approximate functional equation [27,

28]; zeros on the line and simple zeros [3,7,8,14,19,29,34,39,49,55]; the de Bruijn–Newman constant [22,40,48,52], [46, Corollary 2]; Jensen polynomials [16,24,25,47]; J -fractions and Favard’s theorem [17,60].

Computed Quantities

Every other number in the paper is computed from formulas stated in the text by the methods of Section 12: the tower and its waves and energies, the gaps and class counts, the leans and the fitted weights, the half-order limits and the reach of the squared pair, the value distribution and its extremes, the Liouville sums, the Gram matrices and the distances D_N and d_N , the error of the exact weights cut at N , the identity at the first zeros, the zeros of the truncated pair and the Jensen chains, the constants C_{fwd} and C_{rev} , and the readings and densities of the zeros on the prime clocks.

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